Manuscripts considered — ulms-highgrade-pulm-mets-tp53-smarcb1-hla-a2-w9t4

Master inventory: every paper reviewed in this case

Generated 2026-08-25

Manuscripts considered — ulms-highgrade-pulm-mets-tp53-smar

Master inventory: 41 clinical (34 included, 7 considered & excluded) + 53 pre-clinical (47 included, 6 considered & excluded) + 46 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Gounder/Tannir (2026) — Nat Commun

Reference:PMID 41882006 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tazemetostat across SWI/SNF-altered solid tumours (basket, INI1-negative cohort)
- **Indication / model:** solid tumours with SWI/SNF alterations: rhabdoid tumours, synovial sarcoma, INI1-negative tumours of any histology, renal medullary carcinoma, poorly differentiated chordoma (2L+); 129 patients across five cohorts; Cohort 3 (INI1-negative any histology, n=32) is the cohort a tissue-confirmed INI1-lost uLMS would have entered
- **Design:** open-label phase 2 basket, 2-stage Green-Dahlberg design (NCT02601950)
- **n:** 129
- **Effect size:** 9 % — objective response rate, INI1-negative cohort
- **Toxicities:** any treatment-related AE G>=3 (n=129, 10%)
- **Case fit:** weak
- **Notes:** The closest thing to data for an INI1-lost uLMS: a 9% response rate in INI1-negative non-epithelioid histologies, with the authors concluding single-agent EZH2 inhibition needs combination partners. No uLMS-specific EZH2 efficacy data exist anywhere, accrual is closed, and both BRD9 degraders in the SWI/SNF space have been discontinued (see excluded rows). This branch stays hypothetical until tissue INI1 IHC returns lost.

Boieri/Weigand (2026) — Immunother Adv

Reference:PMID 41480010 · **Type:** preclinical · **Inclusion:** excluded — Allogeneic TCR-transduced NK-cell construct rather than either autologous TCR-T product in this case's trial set; the effector biology and the manufacturing constraints both differ enough that it does not inform the afami-cel or lete-cel decision.

- **Intervention:** MAGE-A4-specific TCR-NK cells against solid tumours
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Allogeneic TCR-transduced NK-cell construct rather than either autologous TCR-T product in this case's trial set; the effector biology and the manufacturing constraints both differ enough that it does not inform the afami-cel or lete-cel decision.

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ivonescimab (PD-1 x VEGF bispecific)
- **Indication / model:** unresectable or metastatic leiomyosarcoma (1L or 2L+); —
- **Design:** single-arm phase 2
- **n:** 20

- **Effect size:** — — objective response
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** abemaciclib + gemcitabine and docetaxel
- **Indication / model:** advanced leiomyosarcoma or dedifferentiated liposarcoma (2L+); RB1 / cell-cycle pathway (correlative)
- **Design:** phase 1/2 with randomised phase 2 component
- **n:** 74
- **Effect size:** — — blood thymidine kinase activity time course (phase 1) and progression-free survival (phase 2)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** REC-617 (selective CDK7 inhibitor)
- **Indication / model:** metastatic or unresectable RB1-negative leiomyosarcoma (2L+); RB1 loss (required)
- **Design:** open-label phase 1b
- **n:** 15
- **Effect size:** — — safety and adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ZL-6201 (LRRC15-directed antibody-drug conjugate)
- **Indication / model:** locally advanced or metastatic sarcoma and selected solid tumors (2L+); LRRC15 (sponsor-assessed)
- **Design:** phase 1a/1b dose escalation and expansion
- **n:** 180
- **Effect size:** — — incidence of dose-limiting toxicities
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** valemetostat tosylate (EZH1/2 dual inhibitor)
- **Indication / model:** metastatic or recurrent solid tumors with SWI/SNF loss (Cohort 1.A: SMARCB1-defective) (1L or 2L); bi-allelic (homozygous) SMARCB1 deletion by validated NGS on tissue or plasma, plus centralised IHC confirmation of loss of expression
- **Design:** multi-cohort phase 2 basket
- **n:** 900
- **Effect size:** — — antitumour activity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** abemaciclib + gemcitabine and docetaxel
- **Indication / model:** advanced leiomyosarcoma or dedifferentiated liposarcoma (1L or 2L+ (phase 2 admits systemic-treatment-naive; prior gemcitabine excluded)); intact Rb expression by IHC on baseline or archival tumour (required)
- **Design:** phase 1/2 with randomised phase 2 component
- **n:** 74
- **Effect size:** — — blood thymidine kinase activity time course (phase 1) and progression-free survival (phase 2)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Wermke/Britten (2025) — Nat Med

Reference:PMID 40205198 · **Type:** clinical · **Inclusion:** included

- **Intervention:** IMA203 PRAME-directed TCR T cells (phase 1)
- **Indication / model:** PRAME-positive recurrent/refractory solid tumours in HLA-A*02-positive patients (melanoma-dominant; sarcoma included) (2L+); 41 patients started lymphodepletion, 40 treated; PRAME expression and HLA-A*02 required; confirmed responses enriched at higher dose and higher PRAME expression
- **Design:** first-in-human phase 1 dose escalation/extension (NCT03686124), interim report
- **n:** 41
- **Effect size:** 28.9 % — confirmed objective response rate
- **Toxicities:** cytokine release syndrome G>=3 (2/41, 4.9%); neurotoxicity (ICANS) G>=3 (0/41, 0%)
- **Case fit:** cross_tumor_only
- **Notes:** The published PRAME TCR-T dataset her HLA type points at; melanoma-dominant with sarcoma patients included but no uLMS-specific efficacy reported. PRAME is frequently expressed in sarcomas including LMS, yet her own PRAME expression is untested; the trial screen carries the active IMA203 registration-track study. Brenetafusp, the PRAME ImmTAC, had no peer-reviewed clinical publication as of August 2026.

Xiong/Zhou (2025) — Lancet

Reference:PMID 40057343 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Ivonescimab vs pembrolizumab (HARMONi-2, NSCLC)
- **Indication / model:** PD-L1-positive advanced NSCLC, first line (1L); 398 patients in China; PD-L1 TPS >=1%; EGFR/ALK excluded
- **Design:** randomised double-blind phase 3 (HARMONi-2, NCT05499390), interim analysis
- **n:** 398
- **Effect size:** 0.51 HR — PFS by masked independent review
- **Variance:** 95% CI 0.38–0.69; p=<0.0001
- **Toxicities:** any treatment-related AE G>=3 (58/198, 29%); immune-related AE G>=3 (14/197, 7%)
- **Case fit:** cross_tumor_only
- **Notes:** Why the PD-1xVEGF bispecific is credible: it beat pembrolizumab head-to-head on PFS in NSCLC. Zero published sarcoma efficacy data; the single-arm MSK trial (NCT07516925) is her only route, and the GOG-0250 row is the cautionary precedent that VEGF blockade added to chemo did nothing in uLMS. China-only trial; OS immature at this analysis.

Livingston/Wagner (2025) — Clin Cancer Res

Reference:PMID 39660994 · **Type:** clinical · **Inclusion:** excluded — Program discontinued by the sponsor. Phase 1 in synovial sarcoma and SMARCB1-deficient tumours: ORR 2% (1 of 55), QTc prolongation dose-limiting. Read because a BRD9 degrader was the other clinical route at the SMARCB1 hypothesis; no development path remains.

- **Intervention:** FHD-609 (BRD9 degrader), phase 1
- **Notes:** Excluded: Program discontinued by the sponsor. Phase 1 in synovial sarcoma and SMARCB1-deficient tumours: ORR 2% (1 of 55), QTc prolongation dose-limiting. Read because a BRD9 degrader was the other clinical route at the SMARCB1 hypothesis; no development path remains.

Jackson/Fisher (2025) — J Med Chem

Reference:PMID 41285423 · **Type:** clinical · **Inclusion:** excluded — Program discontinued; monotherapy advancement halted for cardiac toxicity with modest late-line efficacy per the sponsor's own report. Primarily a medicinal-chemistry paper with phase 1 topline; with FHD-609 also stopped, both clinical BRD9 degraders in the SWI/SNF space are gone.

- **Intervention:** CFT8634 (BRD9 degrader), discovery report with phase 1 topline
- **Notes:** Excluded: Program discontinued; monotherapy advancement halted for cardiac toxicity with modest late-line efficacy per the sponsor's own report. Primarily a medicinal-chemistry paper with phase 1 topline; with FHD-609 also stopped, both clinical BRD9 degraders in the SWI/SNF space are gone.

Wang/Yan (2025) — Clin Cancer Res

Reference:PMID 39918552 · **Type:** clinical · **Inclusion:** excluded — Wrong setting: adjuvant therapy for resected localized STS, not metastatic disease. Read while checking the anlotinib evidence base for the phase 3 in the trial screen.

- **Intervention:** Anlotinib adjuvant vs placebo in localized high-grade STS
- **Notes:** Excluded: Wrong setting: adjuvant therapy for resected localized STS, not metastatic disease. Read while checking the anlotinib evidence base for the phase 3 in the trial screen.

Chen/Li (2025) — Cancer Med

Reference:PMID 40152485 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MAGE-A4, PRAME and NY-ESO-1 prevalence across bone and soft-tissue sarcoma
- **Indication / model:** human sarcoma specimens by IHC and multiplex immunostaining microarray; 26 smooth-muscle sarcomas among 128 tumours
- **Design:** cancer/testis antigen expression as the second gate for HLA-restricted TCR and ImmTAC programs
- **n:** n=128 (21 UPS, 26 smooth-muscle sarcoma, 28 liposarcoma, 40 osteosarcoma, 13 chondrosarcoma)
- **Effect size:** weak — Expression was subtype-dependent, highest for MAGE-A4 and PRAME in UPS and osteosarcoma. NY-ESO-1 was low across every subtype, and the smooth-muscle sarcoma group was positive for MAGE-A4 in a minority of cases.
- **Variance:** vs cross-subtype comparison; testis as the reference positive tissue for this antigen class; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The reported MAGE-A4 percentages for smooth-muscle sarcoma are internally inconsistent between the results text and its own framing, and an erratum was issued (Cancer Med 2025;14:e70971) — treat the exact figure as soft. Smooth-muscle sarcoma here is not uterus-restricted.

Guo/Li (2025) — Front Immunol

Reference:PMID 40948800 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Single-cell map of the metastatic uterine leiomyosarcoma microenvironment
- **Indication / model:** single-cell RNA sequencing of metastatic lesions from one treatment-naive uterine leiomyosarcoma patient versus normal myometrium
- **Design:** exhausted CD8 T cells, M2-polarised tumour-associated macrophages and immature N2 neutrophils together with MIF-CD74/CD44 and CXCL8 signalling produce an immunosuppressed niche
- **n:** 1 uLMS patient (4 metastatic sites); n=5 normal myometrium
- **Effect size:** moderate — Metastatic uLMS lesions were dominated by exhausted CD8 T cells, CD163-positive M2 macrophages and CD15-positive immature neutrophils, with naive markers lost along the pseudotime trajectory. CXCL8 signalling linked angiogenesis to macrophage polarisation in the same niche.
- **Variance:** vs normal uterine myometrium; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** A single patient's metastases, so cell-type frequencies are anecdotal even if the qualitative picture matches the wider literature. No therapeutic arm.

Zhong/Li (2025) — iScience

Reference:PMID 40034861 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Ivonescimab tetravalent PD-1 x VEGF bispecific — cooperative binding and Fc silencing
- **Indication / model:** biophysical and cell-based assays plus mouse tumour models; cynomolgus monkey safety
- **Design:** VEGF dimers cross-link the tetravalent antibody into soluble complexes that raise its avidity for PD-1, and PD-1 engagement reciprocally strengthens VEGF binding; Fc mutations abrogate FcgammaRI/IIIa binding
- **Effect size:** moderate — Each target enhanced binding to the other, so PD-1/PD-L1 and VEGF blockade were

both more potent than with the separate antibodies. Fc silencing cut effector function in vitro, consistent with the tolerability seen in monkeys and patients, and the antibody produced a significant in-vivo antitumor response.
- **Variance:** vs parental anti-PD-1 and anti-VEGF antibodies and their combination; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Authored entirely by the sponsor, and the in-vivo work is in humanized-mouse tumour models with no sarcoma among them. Explains the molecular design; says nothing about leiomyosarcoma.

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tislelizumab + liposomal doxorubicin + ifosfamide
- **Indication / model:** advanced soft-tissue sarcoma, first line (1L); —
- **Design:** single-arm phase 2 (TAIS)
- **n:** 45
- **Effect size:** — — objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** all-trans retinoic acid + cemiplimab
- **Indication / model:** advanced or metastatic leiomyosarcoma after standard chemotherapy (2L+); —
- **Design:** single-arm phase 2
- **n:** 16
- **Effect size:** — — overall response rate
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** pegylated liposomal doxorubicin maintenance
- **Indication / model:** advanced soft-tissue sarcoma with disease control after first-line chemotherapy (maintenance after 1L); —
- **Design:** randomised phase 2
- **n:** 81
- **Effect size:** — — progression-free survival from randomisation
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ADCE-D01 (uPARAP/Endo180-directed antibody-drug conjugate)
- **Indication / model:** metastatic or unresectable soft-tissue sarcoma after 1-2 lines (2L+); —
- **Design:** first-in-human phase 1/2
- **n:** 270
- **Effect size:** — — maximum tolerated dose and recommended phase 2 dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** HS-20093 (B7-H3 ADC)
- **Indication / model:** relapsed or refractory osteosarcoma (2L+); B7-H3 (CD276)
- **Design:** randomised phase 3
- **n:** 117
- **Effect size:** — — progression-free survival by independent review
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tebentafusp (gp100 x CD3 ImmTAC)
- **Indication / model:** advanced clear cell sarcoma, HLA-A*02:01 positive (2L+); HLA-A*02:01 positive
- **Design:** randomised phase 2
- **n:** 47
- **Effect size:** — — disease response
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** [225Ac]RTX-2358 (FAP-directed alpha-emitting radioligand), with [64Cu]LNTH-1363S imaging
- **Indication / model:** relapsed and refractory advanced soft-tissue sarcoma (2L+); FAPi PET/CT or PET/MRI uptake in a measurable lesion (required)
- **Design:** phase 1 dose escalation with imaging selection
- **n:** 30

- **Effect size:** — — treatment-emergent adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** [212Pb]Pb-PSV359 (FAP-α-directed alpha-emitting radioligand), gated by [203Pb]Pb-PSV359 imaging
- **Indication / model:** locally advanced or metastatic solid tumors including sarcoma (2L+); [203Pb]Pb-PSV359 SPECT/CT uptake in at least one known lesion at 1 hour
- **Design:** phase 1/2 theranostic pair
- **n:** 112
- **Effect size:** — — safety and tolerability
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Cote GM/Shapiro GI (2025) — Clin Cancer Res

Reference:PMID 39453756 · **Type:** trial · **Inclusion:** included

- **Intervention:** berzosertib (ATR inhibitor) monotherapy, 240 mg/m2 IV twice weekly
- **Indication / model:** molecularly selected advanced solid tumors; cohort T1 was ATRX-mutant leiomyosarcoma (2L+); ATRX mutation by NGS (LMS cohort); ATM, replication-stress genes, SDH in other cohorts
- **Design:** translational phase 2, four molecularly defined cohorts with paired biopsies
- **Effect size:** ATRX-mutant LMS cohort progressed within 4 months; longest disease control was in SDH-mutant GIST (median PFS 229 days) — progression-free survival and pharmacodynamic target engagement
- **Case fit:** weak
- **Notes:** (toxicity table not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tuvusertib (ATR inhibitor) monotherapy
- **Indication / model:** progressive IDH-mutant grade 2-4 astrocytoma with confirmed ATRX and p53 mutation (2L+); ATRX mutation by IHC or NGS plus p53 mutation (required)
- **Design:** single-arm phase 2
- **Effect size:** — — response by RANO 2.0
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Pautier/Duffaud (2024) — N Engl J Med

Reference:PMID 39231341 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Doxorubicin + trabectedin, then trabectedin maintenance (LMS-04, final OS)
- **Indication / model:** metastatic or unresectable leiomyosarcoma (uterine and soft-tissue), chemotherapy-naive (1L); same 150-patient ITT population as the primary report; median follow-up 55 months; analyses adjusted for uterine vs soft-tissue origin and disease stage
- **Design:** randomised open-label phase 3 (LMS-04, NCT02997358), final overall-survival analysis
- **n:** 150
- **Effect size:** 33 months — median overall survival
- **Variance:** 95% CI 26–48
- **Toxicities:** Abstract states AE incidence and dose reductions were higher with the combination; no new per-term rates given. Per-term safety is carried on the primary-report row (pmid 35835135).
- **Case fit:** strong
- **Notes:** The OS confirmation that separates LMS-04 from every other first-line intensification attempt in this disease. Fallback for per-term AEs attempted: abstract has none, NEJM is not in PMC, publisher is paywalled; the trial's per-term safety table is on the companion Lancet Oncol row.

D'Angelo/Van Tine (2024) — Lancet

Reference:PMID 38554725 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Afamitresgene autoleucel (SPEARHEAD-1)
- **Indication / model:** advanced synovial sarcoma or myxoid round cell liposarcoma, HLA-A*02, MAGE-A4-expressing, after anthracycline- or ifosfamide-containing therapy (2L+); 52 patients (44 synovial sarcoma, 8 MRCLS); median 3 prior lines; MAGE-A4 expression and HLA-A*02 required at screening
- **Design:** open-label single-arm phase 2 (SPEARHEAD-1, NCT04044768)
- **n:** 52
- **Effect size:** 37 % — overall response rate by masked independent review
- **Variance:** 95% CI 24–51
- **Toxicities:** cytokine release syndrome (37/52, 71%); cytokine release syndrome G3 (1/52, 2%); lymphopenia G>=3 (50/52, 96%); neutropenia G>=3 (44/52, 85%); leukopenia G>=3 (42/52, 81%)
- **Case fit:** cross_tumor_only
- **Notes:** Basis of the FDA approval (Tecelra), which reaches synovial sarcoma only. She clears the HLA-A*02 gate but sits outside the histology, and her MAGE-A4 status is untested; uLMS access would be trial-based (e.g. surgery-free cohorts or basket TCR-T programs), contingent on antigen IHC.

De Wispelaere/Amant (2024) — Clin Transl Med

Reference:PMID 38711203 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PI3K/mTOR inhibition sensitising pS6-high uterine leiomyosarcoma to PD-1 blockade
- **Indication / model:** humanized and immunodeficient uterine leiomyosarcoma patient-derived xenografts; TCGA analysis of 101 LMS; serial single-cell RNA/TCR sequencing of biopsies
- **Design:** PI3K/mTOR overactivation associates with lymphocyte depletion; inhibiting it repolarises macrophages, raises antigen presentation on dendritic and tumour cells, and draws in PD-1-positive T cells that then need checkpoint release
- **Effect size:** strong — Single-agent anti-PD-1 produced no response in the ICB-resistant humanized uLMS PDX, while combining it with PI3K/mTOR inhibition gave partial or complete responses. The combination

reinvigorated exhausted T cells and drove clonal expansion of a cytotoxic CD8 population supported by CD4 Th1 cells.
- **Variance:** vs single-agent anti-PD-1, single-agent PI3K/mTOR inhibitor, vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Humanized PDX immunology is an approximation of a human immune system, and the pS6-high selection marker has not been measured in this patient. The nearest lineage-matched argument that checkpoint blockade in uLMS needs a partner rather than a companion diagnostic.

Lynch/Pollack (2024) — BMC Cancer

Reference:PMID 39478506 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** B7-H3 expression across soft-tissue sarcoma subtypes
- **Indication / model:** 153 human soft-tissue sarcoma specimens across 15 subtypes, scored by IHC for tumour and vessel staining
- **Design:** B7-H3 surface expression as the antigen available to an ADC or a T-cell engager
- **n:** n=153 patients
- **Effect size:** strong — B7-H3 was expressed in 97% of samples and at high level in 69.2%, with vessel positivity in 94.7%. Expression was independent of prior treatment, tumour size, grade, age, and of PD-L1 or PD-1 status.
- **Variance:** vs within-cohort comparison; PD-L1/PD-1 status where previously assessed; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Prevalence, not density in molecules per cell, which is the quantity that actually governs CAR and ADC activity. Leiomyosarcoma is one of 15 subtypes in the cohort rather than the focus.

Holzmayer/Märklin (2024) — Front Immunol

Reference:PMID 38765008 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CC-3, a B7-H3 x CD3 bispecific antibody, against bone and soft-tissue sarcoma
- **Indication / model:** sarcoma cell lines of multiple subtypes with healthy-donor T cells; patient sarcoma specimens for expression and outcome
- **Design:** IgG-based B7-H3 x CD3 bispecific redirects polyclonal T cells to B7-H3-positive tumour cells independent of TCR specificity or HLA
- **Effect size:** moderate — B7-H3 was found on every sarcoma line tested, and CC-3 triggered T-cell activation, cytokine release, proliferation, memory differentiation and target-restricted lysis. Higher B7-H3 in patient specimens went with shorter progression-free and overall survival.
- **Variance:** vs B7-H3-negative target cells; control bispecific/T cells alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro only — no xenograft arm — and the sarcoma line panel is not leiomyosarcoma-specific. HLA-independence is the practical contrast with the TCR-T options in this case.

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MASCT-I (autologous dendritic cell and cytokine-induced killer cell therapy) + doxorubicin + ifosfamide
- **Indication / model:** advanced soft-tissue sarcoma, first line (1L); —
- **Design:** phase 2
- **n:** 148
- **Effect size:** — — progression-free survival
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** HS-20093 (B7-H3 ADC) + anlotinib, epirubicin, or adebrelimab
- **Indication / model:** advanced bone and soft-tissue sarcoma (1L (dose-expansion) or 2L+); B7-H3 (not required for entry)
- **Design:** phase 1b combination dose escalation and expansion (ARTEMIS-103)
- **n:** 448
- **Effect size:** — — maximum tolerated dose
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** zanzalintinib (XL092)
- **Indication / model:** metastatic or unresectable leiomyosarcoma and other sarcomas (2L+); —
- **Design:** phase 2
- **n:** 73
- **Effect size:** — — progression-free survival
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** gemcitabine + nab-sirolimus
- **Indication / model:** advanced leiomyosarcoma, or sarcoma with TSC1/TSC2 loss (2L+); TSC1/TSC2 loss (alternative entry route)
- **Design:** phase 1 dose finding
- **n:** 18
- **Effect size:** — — safety and adverse events
- **Case fit:** partial

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MGC026 (B7-H3-directed antibody-drug conjugate)
- **Indication / model:** advanced solid tumors including sarcoma (2L+); B7-H3 (CD276), correlative
- **Design:** phase 1 dose escalation and expansion
- **n:** 250
- **Effect size:** — — adverse events and dose-limiting toxicity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GSK5764227 (risvutatug rezetecan, HS-20093), alone or with platinum or checkpoint blockade
- **Indication / model:** advanced or metastatic solid tumors (2L+); B7-H3 (CD276), central testing where tissue available
- **Design:** phase 1a/1b modular study (EMBOLD)
- **n:** 845
- **Effect size:** — — adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CC-3 (CD276 x CD3 bispecific antibody)
- **Indication / model:** advanced colorectal, breast, penile carcinoma and sarcoma (2L+); B7-H3 (CD276)
- **Design:** first-in-human phase 1
- **n:** 89
- **Effect size:** — — safety and adverse events
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ubamatamab (MUC16 x CD3 bispecific), alone or with cemiplimab

- **Indication / model:** MUC16-expressing SMARCB1-deficient renal medullary carcinoma or epithelioid sarcoma (2L+); loss of SMARCB1 by IHC plus serum CA-125 ≥ 70 U/mL or MUC16 IHC H-score > 25
- **Design:** phase 2 with two histology cohorts
- **n:** 15
- **Effect size:** — — safety and adverse events
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CDR404 (MAGE-A4 x CD3 T-cell engager)
- **Indication / model:** selected advanced solid tumors, MAGE-A4-expressing (2L+); HLA-A*02:01 positive plus MAGE-A4-positive tumor
- **Design:** first-in-human phase 1
- **n:** 42
- **Effect size:** — — dose-limiting toxicity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** peposertib (DNA-PK inhibitor) + low-dose pegylated liposomal doxorubicin
- **Indication / model:** metastatic leiomyosarcoma and selected soft-tissue sarcomas (MFS, UPS, synovial, DDLPS) (2L+); —
- **Design:** phase 1 dose escalation with mandatory paired biopsies
- **Effect size:** — — safety and recommended phase 2 dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Ingham/Schwartz (2023) — J Clin Oncol

Reference:PMID 37467452 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib + temozolomide (NCI Protocol 10250)
- **Indication / model:** advanced uterine leiomyosarcoma after progression on >=1 prior line (2L+); 22 evaluable uLMS patients; 59% had >=3 prior lines; HRD not required for entry but assessed on paired biopsies (RAD51 assay, WES/RNAseq)
- **Design:** single-arm multicenter phase 2 (NCI 10250, NCT03880019)
- **n:** 22
- **Effect size:** 23 % — best ORR within 6 months (primary endpoint, met: >=5 of 22 required)
- **Toxicities:** neutropenia G>=3 (n=22, 75%); thrombocytopenia G>=3 (n=22, 32%)

- **Case fit:** partial
- **Notes:** The strongest targeted-combination signal in her exact histology, but it sits one line away: entry required prior therapy. Entry did not require HRD, and half of assayed tumours were RAD51-deficient; her own HRD/BRCA status is unmeasured, which is exactly why tissue testing gates this branch. A randomised phase 2/3 confirmation (NCT05432791) has completed accrual. Published erratum (pmid 37801676) does not change the efficacy results.

Nobori/Kawamura (2023) — JTCVS Open

Reference:PMID 37063124 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pulmonary metastasectomy for uterine malignancies (Japanese multicentre registry)
- **Indication / model:** lung metastases from uterine primaries; sarcoma subset n=46 of 319 (any); Metastatic Lung Tumor Study Group of Japan, 1984-2016; risk-adjusted analysis by primary histology
- **Design:** retrospective multicentre registry analysis
- **n:** 319
- **Effect size:** 55.4 % 5-year OS — 5-year OS after metastasectomy, uterine sarcoma subset
- **Toxicities:** Operative complications not reported in the abstract.
- **Case fit:** partial
- **Notes:** The uterine-specific metastasectomy dataset: half of resected uterine-sarcoma patients alive at 5 years, with DFI over 12 months the main selector. Her DFI qualifies; her current bilateral distribution does not, pending systemic response.

Lebow/Rimner (2023) — Radiother Oncol

Reference:PMID 37532104 · **Type:** clinical · **Inclusion:** included

- **Intervention:** SBRT for sarcoma pulmonary metastases (MSK series)
- **Indication / model:** sarcoma pulmonary metastases, including patients unfit for or declining resection (any); 66 consecutive patients, median follow-up 36 months; oligometastatic and intrathoracic-only disease predicted longer survival
- **Design:** retrospective single-institution cohort
- **n:** 66
- **Effect size:** 7.4 % local failure at 24 months — cumulative incidence of local failure
- **Variance:** 95% CI 4.0–13.9
- **Toxicities:** pneumonitis G1-2 (3/66, 5%); esophagitis G1-2 (1/66, 2%); any adverse event G>=3 (0/66, 0%)
- **Case fit:** partial
- **Notes:** Local control above 90% with essentially no high-grade toxicity makes SBRT the non-surgical consolidation option if induction converts her to oligoprogressive or oligoresidual disease. Baseline pulmonary reserve at 8,500 ft residence should enter any pneumonitis-risk discussion.

Asha/Shah (2023) — Am J Clin Oncol

Reference:PMID 36914598 · **Type:** clinical · **Inclusion:** included

- **Intervention:** SBRT for sarcoma pulmonary metastases (Cleveland Clinic series)
- **Indication / model:** sarcoma lung metastases (any); 50 patients, 109 lung metastases, 2005-2021; lesion size >4 cm predicted worse local control and survival

- **Design:** retrospective single-institution cohort
- **n:** 50
- **Effect size:** 88 % local control at 3 years — local control
- **Toxicities:** any toxicity (8/50, 16%)
- **Case fit:** partial
- **Notes:** Adds the chemotherapy-deferral angle: most patients needed no systemic change for a year after lung SBRT. In her sequencing the same idea would apply only after induction, to consolidate residual disease.

Dall/Barker (2023) — J Exp Clin Cancer Res

Reference:PMID 37143137 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP inhibition in HR-deficient uterine leiomyosarcoma PDX
- **Indication / model:** uterine leiomyosarcoma patient-derived xenografts, including a paired post-PARPi-resistant PDX; national cohort screened by WGS/WES/panel
- **Design:** synthetic lethality of PARP inhibition in tumours with biallelic HR-gene loss; PARP1-selective trapping as the more complete version of that mechanism
- **n:** 58 individuals screened; 5 (9%) HRD; 13 samples by WGS
- **Effect size:** strong — Five of 58 uLMS (9%) were HR-deficient, all through homozygous BRCA2 deletion, and all 13 WGS samples carried a dominant COSMIC signature 3. PDX responses were fastest and most sustained with the PARP1-selective AZD5305, including in a paired PDX that had acquired PRKDC resistance mutations after patient PARPi exposure.
- **Variance:** vs olaparib alone and olaparib plus cisplatin as comparators to the PARP1-selective agent; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The responding models were biallelic BRCA2-deleted, a state this patient has not been tested for; signature 3 was near-universal but CHORD scores were high in only 2 of 13, so genome-wide LOH alone did not equal functional HRD.

Kirkey/Meshinchi (2023) — Blood Adv

Reference:PMID 35984639 · **Type:** preclinical · **Inclusion:** excluded — Model system is AML, where PRAME expression, antigen density and effector access all differ from a solid mesenchymal tumour; contributes nothing on the sarcoma question that Chang 2017 does not cover better.

- **Intervention:** PRAME-directed mTCR-CAR T cells in acute myeloid leukemia
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Model system is AML, where PRAME expression, antigen density and effector access all differ from a solid mesenchymal tumour; contributes nothing on the sarcoma question that Chang 2017 does not cover better.

Brignole/Ponzoni (2023) — J Immunother Cancer

Reference:PMID 37775116 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Vobramitamab duocarmazine in B7-H3-expressing neuroblastoma models
- **Indication / model:** neuroblastoma cell lines and orthotopic/metastatic mouse xenograft models with varying

B7-H3 levels
- **Design:** B7-H3-directed antibody delivering a duocarmycin payload, with membrane-permeable released drug killing neighbouring antigen-low cells
- **Effect size:** strong — The ADC inhibited tumour growth and extended survival across models, with activity extending to cells expressing lower B7-H3 through bystander killing of adjacent cells.
- **Variance:** vs isotype-ADC control; untreated mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Neuroblastoma only, and the payload class differs from the topoisomerase-I payloads carried by the B7-H3 ADCs in this case's trial set. The bystander principle is what transfers, not the potency numbers.

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** lurbinectedin + doxorubicin
- **Indication / model:** metastatic or unresectable leiomyosarcoma, first line (1L); —
- **Design:** randomised phase 3 (LMS-05)
- **n:** 469
- **Effect size:** — — progression-free survival by independent review
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** olaparib + temozolomide
- **Indication / model:** advanced uterine leiomyosarcoma after prior therapy (2L+); —
- **Design:** randomised phase 2/3
- **n:** 74
- **Effect size:** — — progression-free survival
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** fluzoparib + camrelizumab with concurrent stereotactic body radiotherapy
- **Indication / model:** metastatic or advanced high-grade bone and soft-tissue sarcoma (2L+); —
- **Design:** single-arm phase 2
- **n:** 86
- **Effect size:** — — 6-month progression-free survival rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** [177Lu]Lu-ITG-PSMA-1, selected by [68Ga]Ga-PSMA-11 PET/CT
- **Indication / model:** soft-tissue sarcoma progressing after standard therapy (2L+); [68Ga]Ga-PSMA-11 PET/CT uptake in tumour neovasculature
- **Design:** phase 1 theranostic, vascular-disruption approach
- **n:** 14
- **Effect size:** — — imaging result on [68Ga]Ga-PSMA-11 PET/CT
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tuvusertib (ATR inhibitor) + temozolomide
- **Indication / model:** advanced solid tumors; dose escalation admits soft-tissue sarcoma as a temozolomide standard-of-care histology (2L+); MGMT promoter hypermethylation, or a histology where temozolomide is standard per guidelines (soft-tissue sarcoma qualifies)
- **Design:** phase 1/2, dose escalation then MSS colorectal phase 2
- **Effect size:** — — safety and recommended phase 2 dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Pautier/Duffaud (2022) — Lancet Oncol

Reference:PMID 35835135 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Doxorubicin + trabectedin, then trabectedin maintenance (LMS-04)
- **Indication / model:** metastatic or unresectable leiomyosarcoma (uterine and soft-tissue), chemotherapy-naive (1L); ECOG 0-1; 67 of 150 patients had uterine LMS; randomisation stratified by uterine vs soft-tissue origin; surgery for residual disease allowed after 6 cycles
- **Design:** randomised open-label phase 3 (LMS-04, NCT02997358)
- **n:** 150
- **Effect size:** 12.2 months — median PFS by blinded independent central review
- **Variance:** 95% CI 10.1–15.6; p=<0.0001
- **Toxicities:** neutropenia G>=3 (59/74, 80%); thrombocytopenia G>=3 (35/74, 47%); anemia G>=3 (23/74, 31%); febrile neutropenia G>=3 (21/74, 28%); treatment-related death G5 (1/150, 0.7%)
- **Case fit:** strong
- **Notes:** The regimen her oncologist has planned. Uterine LMS was 45% of the trial and a stratification factor, but the abstract does not report a uterine-only PFS estimate; the benefit claim for her rests on the stratified full-trial analysis. AE rates from the published abstract.

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tiragolumab + atezolizumab
- **Indication / model:** relapsed or refractory SMARCB1- or SMARCA4-deficient tumors (2L+); SMARCB1 (INI1) or SMARCA4 deficiency by institutional IHC, or molecular confirmation where IHC is equivocal
- **Design:** phase 1/2 with adult phase 2 cohorts
- **n:** 86
- **Effect size:** — — dose-limiting toxicity and response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CFT8634 (BRD9 degrader)
- **Indication / model:** locally advanced or metastatic SMARCB1-perturbed solid tumors (2L+); SMARCB1-null tumor
- **Design:** phase 1
- **n:** 49
- **Effect size:** — — frequency and severity of adverse events
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** berzosertib + gemcitabine
- **Indication / model:** metastatic or unresectable leiomyosarcoma, up to 3 prior lines (2L+); —
- **Design:** randomised phase 2, gemcitabine with or without berzosertib
- **Effect size:** — — progression-free survival
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Nathan/Piperno-Neumann (2021) — N Engl J Med

Reference:PMID 34551229 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tebentafusp (ImmTAC class precedent, IMCgp100-202)
- **Indication / model:** previously untreated metastatic uveal melanoma, HLA-A*02:01-positive (1L); 378 HLA-A* 02:01 patients randomised 2:1 vs investigator's choice (pembrolizumab, ipilimumab, or dacarbazine)
- **Design:** randomised open-label phase 3 (NCT03070392)

- **n:** 378
- **Effect size:** 0.51 HR — overall survival
- **Variance:** 95% CI 0.37–0.71; p=<0.001
- **Toxicities:** rash (n=252, 83%); pyrexia (n=252, 76%); pruritus (n=252, 69%); treatment discontinuation for AE (n=252, 2%)
- **Case fit:** cross_tumor_only
- **Notes:** Included as the proof that an off-the-shelf HLA-A*02:01-restricted bispecific TCR can move overall survival in a solid tumour. Everything else about it is uveal-melanoma-specific (gp100 target); its role in her dossier is to anchor the ImmTAC class, not to suggest tebentafusp itself.

Kihara/Niki (2021) — Hum Pathol

Reference:PMID 34271067 · **Type:** clinical · **Inclusion:** excluded — Not an intervention study; IHC prevalence series. Read to weigh the SMARCB1 branch: 170 of 170 uterine smooth-muscle tumours retained SMARCB1 expression, and all 206 uterine mesenchymal tumours retained SMARCB1. This is the hard prior against her plasma-only SMARCB1 frameshift being a tumour feature, and it is why tissue INI1 IHC gates the entire EZH2 branch.

- **Intervention:** SMARCB1/SMARCA4/SMARCA2 IHC in uterine mesenchymal tumours (context for EZH2 branch)
- **Notes:** Excluded: Not an intervention study; IHC prevalence series. Read to weigh the SMARCB1 branch: 170 of 170 uterine smooth-muscle tumours retained SMARCB1 expression, and all 206 uterine mesenchymal tumours retained SMARCB1. This is the hard prior against her plasma-only SMARCB1 frameshift being a tumour feature, and it is why tissue INI1 IHC gates the entire EZH2 branch.

Kihara/Niki (2021) — Hum Pathol

Reference:PMID 34271067 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** SMARCB1/INI1 expression across uterine mesenchymal tumours
- **Indication / model:** human surgical specimens, 206 uterine mesenchymal tumours (170 smooth-muscle tumours) by IHC
- **Design:** INI1 protein readout of SMARCB1 status in the exact anatomic lineage in question
- **n:** n=206 tumours; 170 smooth-muscle
- **Effect size:** negative — Every one of the 206 uterine mesenchymal tumours retained SMARCB1 expression, including all 170 smooth-muscle tumours. Two tumours lost SMARCA4/SMARCA2, and neither was a leiomyosarcoma.
- **Variance:** vs internal non-neoplastic stroma/endothelium as staining control; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Single-institution Japanese cohort scored by IHC alone, so a subclonal or metastasis-restricted SMARCB1 event would not be captured. This is the hardest available prior against INI1 loss in this histology and it is a prior, not a result for this patient.

Alpsoy/Dykhuizen (2021) — Cancer Res

Reference:PMID 33355184 · **Type:** preclinical · **Inclusion:** excluded — BRD9 dependency here is androgen-receptor-driven in prostate epithelium, not a consequence of cBAF perturbation, so the mechanism diverges from the SMARCB1-loss rationale that is at issue in this case.

- **Intervention:** BRD9 in androgen-receptor signalling and prostate cancer
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: BRD9 dependency here is androgen-receptor-driven in prostate epithelium, not a consequence of cBAF perturbation, so the mechanism diverges from the SMARCB1-loss rationale that is at issue in this case.

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** trabectedin
- **Indication / model:** metastatic or locally advanced leiomyosarcoma after anthracycline (2L+); —
- **Design:** randomised phase 2 with observational cohort
- **n:** 100
- **Effect size:** — — growth modulation index
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FHD-609 (BRD9 degrader)
- **Indication / model:** advanced synovial sarcoma or SMARCB1-loss tumors (1L or 2L+); bi-allelic SMARCB1 alteration or corresponding protein loss
- **Design:** phase 1 dose escalation
- **n:** 55
- **Effect size:** — — treatment-emergent adverse events
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** [177Lu]Lu-FAP-2286, selected by [68Ga]Ga-FAP-2286 PET
- **Indication / model:** advanced solid tumors with FAP-2286 uptake (2L+); [68Ga]Ga-FAP-2286 PET uptake (required)
- **Design:** phase 1/2 theranostic (LuMIERE)
- **n:** 222
- **Effect size:** — — dose-limiting toxicity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Gounder/Stacchiotti (2020) — Lancet Oncol

Reference:PMID 33035459 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tazemetostat in INI1/SMARCB1-lost epithelioid sarcoma (approval cohort)
- **Indication / model:** advanced epithelioid sarcoma with INI1/SMARCB1 loss (any); 62 patients, ECOG 0-2, documented INI1 loss by IHC or biallelic SMARCB1 alteration; 1L and later lines
- **Design:** open-label phase 2 basket, epithelioid sarcoma cohort (NCT02601950)
- **n:** 62
- **Effect size:** 15 % — investigator-assessed objective response rate
- **Variance:** 95% CI 7–26
- **Toxicities:** anemia G>=3 (4/62, 6%); weight loss G>=3 (2/62, 3%); treatment-related serious AE G>=3 (2/62, 3%)
- **Case fit:** cross_tumor_only
- **Notes:** Pivotal cohort behind the FDA accelerated approval, which is confined to epithelioid sarcoma. For her this is mechanism precedent only: a different histology, and her SMARCB1 call is plasma-only against a strong prior that uterine smooth-muscle tumours retain INI1 (see excluded Kihara row). Any use in uLMS would be off-label extrapolation gated on tissue INI1 loss.

Pollack/Cranmer (2020) — JAMA Oncol

Reference:PMID 32910151 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Doxorubicin + pembrolizumab, first-line chemo-IO (phase 1/2)
- **Indication / model:** advanced anthracycline-naive sarcoma (LMS the most common subtype, 11 of 37) (1L); 37 patients, single academic sarcoma centre; osteosarcoma, Ewing and alveolar/embryonal rhabdomyosarcoma excluded
- **Design:** non-randomised phase 1/2 (NCT02888665)
- **n:** 37
- **Effect size:** 19 % — objective response rate (primary endpoint not met)
- **Toxicities:** No dose-limiting toxicities or unexpected safety signals; tolerability of the chemo-IO combination was the main positive finding.
- **Case fit:** partial
- **Notes:** Shows anthracycline plus PD-1 is deliverable first line in sarcoma with PFS/OS that compare well with historical doxorubicin, though the response endpoint failed and durable responses clustered in UPS/DDLPS rather than LMS. Context for weighing chemo-first vs the ivonescimab seat. AE fallback: abstract and PMC text report no per-term G3 table beyond absence of DLTs.

Marabelle/Bang (2020) — Lancet Oncol

Reference:PMID 32919526 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab in TMB-high solid tumours (KEYNOTE-158 prospective TMB analysis)
- **Indication / model:** previously treated advanced solid tumours across ten cohorts (no sarcoma cohort) (2L+); 790 efficacy-evaluable, 102 tTMB-high (>=10 mut/Mb by FoundationOne CDx)
- **Design:** prospective biomarker analysis of a multicohort single-arm phase 2 (NCT02628067)
- **n:** 790
- **Effect size:** 29 % — ORR in tTMB-high vs non-tTMB-high

- **Variance:** 95% CI 21–39
- **Toxicities:** any treatment-related AE G>=3 (16/105, 15%); colitis G>=3 (2/105, 2%); pneumonia G5 (1/105, 1%)
- **Case fit:** cross_tumor_only
- **Notes:** Underpins the tumour-agnostic TMB>=10 pembrolizumab approval. Her TMB is unmeasured and uLMS is rarely TMB-high, so this row exists to justify closing the TMB gap on tissue NGS, not to predict benefit. Sarcoma was not among the ten cohorts.

Laroche-Clary/Italiano (2020) — Sci Rep

Reference:PMID 32366852 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibition plus gemcitabine in soft-tissue sarcoma, independent of ALT
- **Indication / model:** soft-tissue sarcoma cell lines and an undifferentiated pleomorphic sarcoma patient-derived xenograft
- **Design:** ATR inhibition removes the intra-S-phase checkpoint that would otherwise let cells survive gemcitabine-induced replication stress
- **Effect size:** moderate — The VE-822 and gemcitabine combination was synergistic in vitro with increased gammaH2AX and S-phase accumulation, and in vivo it improved tumour growth inhibition and progression-free survival over either agent. Sensitivity did not track ALT status.
- **Variance:** vs single-agent VE-822, single-agent gemcitabine, vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The in-vivo model was undifferentiated pleomorphic sarcoma, not leiomyosarcoma. Supports ATR inhibition as a chemo-sensitiser rather than as an ALT-selected strategy.

Goncalves/Tomita (2020) — ACS Pharmacol Transl Sci

Reference:PMID 33344901 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Telomere length, not ALT status, tracked ATR-inhibitor sensitivity
- **Indication / model:** osteosarcoma cell lines of varying telomere length and telomere-maintenance mechanism
- **Design:** short telomeres, rather than the recombination-based maintenance mechanism itself, create the ATR dependency
- **Effect size:** moderate — ATR inhibitors selectively eliminated osteosarcoma lines with short telomeres regardless of whether those lines used ALT.
- **Variance:** vs long-telomere lines within the same panel; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Osteosarcoma lines, in vitro. Reframes the biomarker as telomere length, which nobody has measured in this patient either.

Sanderson/Gerry (2020) — Oncoimmunology

Reference:PMID 32002290 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ADP-A2M4 (afamitresgene) affinity-enhanced MAGE-A4 TCR — preclinical potency and specificity
- **Indication / model:** 2D and 3D human cell cultures, primary tumour material, and human primary-cell and

cell-line cross-reactivity panels
- **Design:** affinity-enhanced TCR recognising the HLA-A*02-restricted MAGE-A4 peptide GVYDGREHTV on the tumour cell surface
- **Effect size:** strong — ADP-A2M4 killed MAGE-A4-positive HLA-A2-positive targets potently across 2D, 3D and primary tumour material. Alanine-scanning and primary-cell screening turned up no major off-target cross-reactivity.
- **Variance:** vs non-transduced T cells; HLA-A2-negative and MAGE-A4-negative targets; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Adaptimmune-authored; the tumour material tested was not uterine leiomyosarcoma. Potency is contingent on antigen expression, which is exactly what has not been measured in this patient.

Petitprez/Fridman (2020) — Nature

Reference:PMID 31942077 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Sarcoma immune classes and B-cell-rich tertiary lymphoid structures
- **Indication / model:** gene-expression profiling of 608 soft-tissue sarcomas with validation against a phase 2 pembrolizumab cohort
- **Design:** composition of the tumour microenvironment, and specifically B-cell-rich tertiary lymphoid structures, governs whether PD-1 blockade has anything to act on
- **n:** n=608 tumours
- **Effect size:** strong — Five immune phenotypes emerged, from immune-low (A, B) to immune-high (D, E). Class E, defined by tertiary lymphoid structures rich in B cells and follicular dendritic cells, had better survival and a high response rate to pembrolizumab, and B-cell content outperformed CD8 content as a predictor.
- **Variance:** vs cross-class comparison within the same cohort; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Leiomyosarcoma sits predominantly in the immune-low classes, which is the structural reason single-agent checkpoint blockade underperforms in this histology. Class assignment requires an expression assay nobody has run here.

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** brenetafusp (IMC-F106C, PRAME x CD3 ImmTAC)
- **Indication / model:** selected advanced solid tumors, PRAME-positive (2L+); HLA-A*02:01 positive plus PRAME-positive tumor
- **Design:** phase 1/2, monotherapy and checkpoint-inhibitor combinations
- **n:** 410
- **Effect size:** — — dose-limiting toxicity and response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Hong DS/Drilon A (2020) — Lancet Oncol

Reference:PMID 32105622 · **Type:** trial · **Inclusion:** included

- **Intervention:** larotrectinib
- **Indication / model:** TRK fusion-positive solid tumors, any histology (1L or 2L+); NTRK1, NTRK2 or NTRK3 gene fusion
- **Design:** pooled analysis of three phase 1/2 trials
- **n:** 159
- **Effect size:** 79% ORR — objective response rate
- **Variance:** 95% CI 72–85
- **Case fit:** weak
- **Notes:** (toxicity table not extracted)

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** pulmonary suffusion with cisplatin plus metastasectomy
- **Indication / model:** sarcoma metastatic to the lungs (n/a (locoregional)); —
- **Design:** phase 1/2 of pulmonary suffusion with metastasectomy
- **n:** 99
- **Effect size:** — — incidence of local toxicity
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Seligson/Chen (2019) — Oncologist

Reference:PMID 30541756 · **Type:** clinical · **Inclusion:** included

- **Intervention:** PARP inhibition in BRCA2-altered uterine LMS (genomic series)
- **Indication / model:** leiomyosarcoma; BRCA2-altered uterine LMS subset (any); 2,548 STS genomic profiles across two cohorts; BRCA2 alterations concentrated in uLMS (10% of uLMS); 4 uLMS patients with functional BRCA2 loss treated with PARP inhibitors
- **Design:** retrospective genomic analysis plus 4-patient case series
- **n:** 4
- **Effect size:** 4 of 4 with durable clinical benefit patients — durable clinical benefit on PARP inhibition (descriptive)
- **Toxicities:** No systematic AE reporting; four-patient off-label series.
- **Case fit:** partial
- **Notes:** Why BRCA2 testing earns its place on her tissue-NGS order: the one STS subtype where BRCA2 loss clusters is hers. Hypothesis-generating only; the prospective data live on the NCI 10250 row. AE data not reported in source beyond case narratives.

Stacchiotti/Zaffaroni (2019) — Cancers (Basel)

Reference:PMID 31331120 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EZH2 inhibition versus conventional chemotherapy in an INI1-negative epithelioid sarcoma PDX
- **Indication / model:** INI1-deficient proximal-type epithelioid sarcoma patient-derived xenograft (PDX ES-1) and

the derived ES-1 cell line
- **Design:** EZH2 methyltransferase inhibition in a tumour with complete INI1 loss, with autophagy induction as the escape route
- **Effect size:** moderate — Maximum tumour volume inhibition of roughly 90% was reached by gemcitabine, by EPZ-011989 and by doxorubicin plus ifosfamide alike. RNA-seq and functional work on residual tumour pointed to autophagy, with HMGA2 central, as a cytoprotective response to EZH2 inhibition.
- **Variance:** vs untreated mice; doxorubicin, ifosfamide, doxorubicin+ifosfamide and gemcitabine arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** One PDX from one INI1-negative epithelioid sarcoma. EZH2 inhibition matched rather than beat anthracycline-based chemotherapy in this model, which is worth holding next to any cross-histology extrapolation.

Wang/Roberts (2019) — Nat Commun

Reference:PMID 31015438 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** BRD9 vulnerability in SMARCB1-mutant rhabdoid tumours and why the bromodomain is the wrong handle
- **Indication / model:** genome-wide CRISPR-Cas9 screen across cancer lines with follow-up in SMARCB1-mutant rhabdoid tumour lines
- **Design:** SMARCB1 loss increases BRD9 incorporation into a distinct SWI/SNF sub-complex whose integrity depends on the DUF3512 domain
- **Effect size:** strong — BRD9 scored as a specific dependency in rhabdoid tumours, and SMARCB1 loss itself drove more BRD9 into the complex. The bromodomain was dispensable while DUF3512 was essential, which is the mechanistic argument for degrading BRD9 rather than inhibiting its bromodomain.
- **Variance:** vs SMARCB1-wild-type lines; domain-deletion BRD9 rescue constructs; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Paediatric rhabdoid lines with biallelic SMARCB1 loss throughout; nothing on heterozygous alleles or mesenchymal smooth-muscle lineage. An erratum was published (Nat Commun 2019;10:4445).

Majzner/Mackall (2019) — Clin Cancer Res

Reference:PMID 30655315 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** B7-H3 CAR T cells and the antigen-density threshold
- **Indication / model:** xenograft models of osteosarcoma, medulloblastoma and Ewing sarcoma; target lines titrated for surface B7-H3
- **Design:** CAR engagement scales with surface antigen copy number, so a density threshold separates tumour killing from sparing of low-expressing normal tissue
- **Effect size:** strong — B7-H3 CAR T cells regressed established solid-tumour xenografts, and efficacy depended heavily on high surface antigen density. Activity fell away sharply against low-expressing targets, which is what creates the therapeutic window over normal tissue.
- **Variance:** vs untransduced T cells; antigen-low and antigen-negative target lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only

- **Notes:** Paediatric tumour xenografts, no leiomyosarcoma. The density-threshold result is the reason a 'B7-H3 positive' IHC call is insufficient on its own for a CAR strategy, and it applies with less force to an ADC with a bystander payload.

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMA203 / IMA203CD8 (PRAME-directed TCR-engineered T cells), alone or with nivolumab
- **Indication / model:** recurrent or refractory PRAME-positive advanced solid tumors (2L+); HLA-A*02:01 positive plus PRAME expression by the sponsor's IMADetect RT-qPCR assay
- **Design:** phase 1/2 dose escalation and expansion (ACTengine)
- **n:** 375
- **Effect size:** — — maximum tolerated dose and objective response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** letetresgene autoleucel (lete-cel, NY-ESO-1 TCR-engineered T cells)
- **Indication / model:** NY-ESO-1-positive synovial sarcoma or myxoid/round-cell liposarcoma (2L+); HLA-A*02:01, A*02:05 or A*02:06 positive plus NY-ESO-1 expression by central laboratory
- **Design:** master protocol, phase 2 substudies (IGNYTE-ESO)
- **n:** 103
- **Effect size:** — — overall response rate
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** uzatresgene autoleucel (ADP-A2M4CD8), alone or with nivolumab or pembrolizumab
- **Indication / model:** HLA-A*02-positive MAGE-A4-expressing carcinomas including endometrial cancer (2L+); at least one HLA-A*02 inclusion allele plus MAGE-A4 expression by central laboratory
- **Design:** phase 1 dose escalation and expansion (SURPASS)
- **n:** 120
- **Effect size:** — — safety and tolerability
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

D'Angelo/Mackall (2018) — Cancer Discov

Reference:PMID 29891538 · **Type:** clinical · **Inclusion:** included

- **Intervention:** NY-ESO-1c259 TCR T cells (letetresgene autoleucel lineage) in synovial sarcoma
- **Indication / model:** metastatic synovial sarcoma, HLA-A*02, NY-ESO-1-expressing (2L+); 12 patients in the reported high-antigen, full-lymphodepletion cohort
- **Design:** single-arm pilot (phase 1/2)
- **n:** 12
- **Effect size:** 50 % — confirmed objective response rate
- **Toxicities:** cytokine release syndrome (5/12, 42%); lymphopenia G>=3 (n=12, 100%); leukopenia G>=3 (n=12, 92%); neutropenia G>=3 (n=12, 83%); anemia G>=3 (n=12, 83%); febrile neutropenia G>=3 (n=12, 17%)
- **Case fit:** cross_tumor_only
- **Notes:** The clinical foundation for lete-cel, which uses this same NY-ESO-1c259 TCR. The registrational IGNYTE-ESO synovial sarcoma/MRCLS results had no PubMed-indexed primary publication as of August 2026, so this pilot is the citable evidence. No uLMS data exist for NY-ESO-1 TCR-T; her NY-ESO-1 expression is untested. CRS and cytopenia rates from the PMC full text.

D'Angelo/Streicher (2018) — Lancet Oncol

Reference:PMID 29370992 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Nivolumab +/- ipilimumab (Alliance A091401)
- **Indication / model:** metastatic sarcoma after >=1 prior systemic line (2L+); 85 treated across two non-comparative arms; LMS among enrolled histologies; responses in the combination arm included LMS and uterine sarcoma
- **Design:** randomised non-comparative phase 2 x2 (NCT02500797)
- **n:** 85
- **Effect size:** 16 % — confirmed ORR per arm
- **Variance:** 95% CI 7–30
- **Toxicities:** anemia G>=3 (8/42, 19%); hypotension G>=3 (4/42, 10%); treatment-related serious AE (11/42, 26%)
- **Case fit:** partial
- **Notes:** Establishes that dual checkpoint blockade retrieves a modest response rate in sarcoma where PD-1 alone fails; the authors sent the combination, not the monotherapy, forward. Supports combination or bispecific IO logic in her disease more than any single-agent plan.

Chi/Cai (2018) — Clin Cancer Res

Reference:PMID 29895706 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Anlotinib in refractory STS (phase 2, LMS subset)
- **Indication / model:** refractory metastatic soft-tissue sarcoma after anthracycline, angiogenesis-inhibitor-naive (2L+); 166 patients in China; LMS subset n=26, not uterine-specific
- **Design:** single-arm phase 2
- **n:** 166
- **Effect size:** 13 % — objective response rate
- **Variance:** 95% CI 7.6–18
- **Toxicities:** hypertension G>=3 (n=166, 4.8%); triglyceride elevation G>=3 (n=166, 3.6%); pneumothorax G>=3 (n=166, 2.4%)

- **Case fit:** weak
- **Notes:** Backs the anlotinib phase 3 in the trial screen: the 26-patient LMS subset did unusually well (11-month mPFS). Chinese population, subgroup not prespecified for inference, and no uterine breakdown; pneumothorax risk deserves attention given her lung disease.

Lindsay/Gibbs (2018) — Sarcoma

Reference:PMID 29808081 · **Type:** clinical · **Inclusion:** excluded — Superseded for this dossier by two larger, more recent sarcoma-lung-SBRT series in the same population (Lebow 2023, Asha 2023) that carry the included rows; findings concordant (95% lesion control).

- **Intervention:** SBRT for sarcoma lung metastases (Florida series)
- **Notes:** Excluded: Superseded for this dossier by two larger, more recent sarcoma-lung-SBRT series in the same population (Lebow 2023, Asha 2023) that carry the included rows; findings concordant (95% lesion control).

Zhang/Huang (2018) — Sci Rep

Reference:PMID 30120321 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EZH2 protein levels in leiomyosarcoma versus leiomyoma
- **Indication / model:** human myogenic tumour specimens by IHC (leiomyosarcoma vs leiomyoma, rhabdomyosarcoma vs rhabdomyoma) with GEO expression datasets
- **Design:** PRC2 subunit expression as a correlate of malignancy in smooth-muscle tumours
- **Effect size:** weak — EZH2 protein was high in leiomyosarcoma and low or absent in leiomyoma, with SUZ12 and EED following a weaker version of the same pattern.
- **Variance:** vs benign myogenic counterparts; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Expression is not dependency. High EZH2 in an INI1-intact smooth-muscle tumour is a proliferation correlate, and none of these tumours were tested against an EZH2 inhibitor.

Michel/Kadoch (2018) — Nat Cell Biol

Reference:PMID 30397315 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ncBAF/BRD9 as a synthetic-lethal target in BAF-perturbed cancers
- **Indication / model:** synovial sarcoma and malignant rhabdoid tumour cell lines; chemical (BRD9 degrader/inhibitor) and shRNA depletion; genome-wide complex mapping
- **Design:** ncBAF complexes localise to CTCF sites and promoters and sustain gene expression when cBAF is crippled by SMARCB1 loss or SS18-SSX fusion
- **Effect size:** strong — Depleting BRD9 rapidly cut proliferation of synovial sarcoma and rhabdoid lines but spared cBAF-intact cancers. The surviving dependency in SMARCB1-perturbed cells ran through ncBAF maintenance of retained CTCF-promoter sites.
- **Variance:** vs cBAF-intact cancer lines; non-targeting constructs; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Cell-line dependency mapping with limited in-vivo work; both index histologies carry complete cBAF

perturbation. This is the rationale that sent FHD-609 and CFT8634 into the clinic, and both programmes were later discontinued.

Brien/Armstrong (2018) — Elife

Reference:PMID 30431433 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** dBRD9 degradation reverses the oncogenic program in synovial sarcoma
- **Indication / model:** synovial sarcoma cell lines treated with the BRD9 degrader dBRD9; proteomics and expression profiling
- **Design:** targeted degradation removes BRD9 protein and the ncBAF assembly it scaffolds, undoing SS18-SSX-driven transcription
- **Effect size:** strong — Degrading BRD9 reversed the oncogenic expression program and suppressed synovial sarcoma growth, where bromodomain occupancy alone was insufficient.
- **Variance:** vs bromodomain-inhibitor comparator and non-SS18-SSX lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Synovial sarcoma is driven by SS18-SSX-mediated SMARCB1 ejection, a different route to the same complex defect than a SMARCB1 mutation; largely in-vitro.

Chudasama/Fröhling (2018) — Nat Commun

Reference:PMID 29321523 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Leiomyosarcoma genomic and transcriptomic landscape
- **Indication / model:** human leiomyosarcoma tumours profiled by whole-exome/genome and RNA sequencing, with functional follow-up in LMS cell lines
- **Design:** recurrent inactivation of TP53, RB1 and PTEN with widespread copy-number change; telomere maintenance and DNA-damage-response features rather than a single driver oncogene
- **Effect size:** Leiomyosarcoma is copy-number driven, with TP53 and RB1 loss recurring and few actionable point mutations. Telomere-maintenance and homologous-recombination features stood out as the recurring biology.
- **Variance:** vs normal tissue comparators for somatic calling; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Mixed anatomic sites rather than uterus-only. Sets the expectation that a TP53 nonsense call in this histology is background biology rather than a lead.

Seddon/Beare (2017) — Lancet Oncol

Reference:PMID 28882536 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + docetaxel (GeDDiS)
- **Indication / model:** previously untreated advanced or metastatic soft-tissue sarcoma (1L); Trojani grade 2-3 STS; leiomyosarcoma (uterine and non-uterine) was the largest histology stratum
- **Design:** randomised controlled phase 3 (GeDDiS, ISRCTN07742377)
- **n:** 257
- **Effect size:** 46.4 % progression-free at 24 weeks — proportion alive and progression-free at 24 weeks (primary

endpoint: no difference)
- **Variance:** 95% CI 37.5–54.8; p=0.06
- **Toxicities:** neutropenia G>=3 (25/126, 20%); febrile neutropenia G>=3 (15/126, 12%); fatigue G>=3 (17/126, 14%); oral mucositis G>=3 (2/126, 2%)
- **Case fit:** partial
- **Notes:** The trial that kept doxorubicin as the default first line: gem-doce was not better and trended worse on PFS. Relevant to her as the strongest randomised argument against choosing gem-doce over an anthracycline backbone up front, GOG-0250's uterine-specific single-arm activity notwithstanding.

Hensley/Demetri (2017) — Gynecol Oncol

Reference:PMID 28651804 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Trabectedin vs dacarbazine, uterine LMS subgroup
- **Indication / model:** advanced uterine leiomyosarcoma after anthracycline failure (2L+); 232 uLMS patients (trabectedin 144, dacarbazine 88), the largest subgroup (40%) of the phase 3 trial; post hoc analysis
- **Design:** post hoc subgroup analysis of a randomised phase 3 (NCT01343277)
- **n:** 232
- **Effect size:** 4.0 months — median PFS in uLMS
- **Variance:** p=0.0012
- **Toxicities:** ALT elevation G>=3 (47/140, 34%); neutropenia G>=3 (51/140, 36%); leukopenia G>=3 (36/140, 26%); anemia G>=3 (28/140, 20%); thrombocytopenia G>=3 (21/140, 15%)
- **Case fit:** partial
- **Notes:** The uterine-specific read of SAR-3007: PFS benefit holds in her histology, response rates stay near 10%, and OS does not move. Post hoc, so treat the subgroup HR as estimation rather than confirmation. Per-term AE rates from the PMC full text.

Tawbi/Patel (2017) — Lancet Oncol

Reference:PMID 28988646 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab monotherapy in STS (SARC028) - LMS cohort negative
- **Indication / model:** advanced soft-tissue and bone sarcoma, up to 3 prior lines (2L+); STS cohort n=40 evaluable, 10 per histology including 10 LMS
- **Design:** two-cohort single-arm phase 2 (SARC028, NCT02301039)
- **n:** 80
- **Effect size:** 18 % — objective response, STS cohort (primary endpoint not met)
- **Toxicities:** anemia G>=3 (3/42, 7%); lymphopenia G>=3 (3/42, 7%); pneumonitis (immune-related) G>=3 (2/84, 2%)
- **Case fit:** partial
- **Notes:** Half of the case against single-agent PD-1 in her histology: zero of ten LMS responses, while UPS and dedifferentiated liposarcoma carried the signal.

Ben-Ami/George (2017) — Cancer

Reference:PMID 28440953 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Nivolumab monotherapy in uterine LMS (negative phase 2)

- **Indication / model:** advanced uterine leiomyosarcoma, previously treated (2L+); 12 patients, single centre; stage 1 of a 2-stage design; PD-L1 expression 20% of archival samples
- **Design:** single-arm phase 2, stopped at stage 1 for futility
- **n:** 12
- **Effect size:** 0 % — objective response rate
- **Toxicities:** amylase/lipase increase G4 (1/12, 8%); fatigue G3 (1/12, 8%); abdominal pain G3 (1/12, 8%)
- **Case fit:** partial
- **Notes:** The other half of the case: the only uLMS-specific single-agent PD-1 trial, stopped for futility at 12 patients with no responses and 1.8-month PFS. Any checkpoint strategy for her needs a combination rationale. AE detail from the PMC full text.

Demetri/Chawla (2017) — J Clin Oncol

Reference:PMID 28854066 · **Type:** clinical · **Inclusion:** excluded — Liposarcoma subgroup analysis; outside her histology. Read to check whether the phase 3 OS benefit generalises to LMS: it was concentrated in liposarcoma (OS HR 0.51), so the trial-level advantage should not be projected onto her. Caveat carried on the eribulin trial row.

- **Intervention:** Eribulin vs dacarbazine, liposarcoma subgroup
- **Notes:** Excluded: Liposarcoma subgroup analysis; outside her histology. Read to check whether the phase 3 OS benefit generalises to LMS: it was concentrated in liposarcoma (OS HR 0.51), so the trial-level advantage should not be projected onto her. Caveat carried on the eribulin trial row.

Nakayama/Kadoch (2017) — Nat Genet

Reference:PMID 28945250 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** SMARCB1 as the BAF subunit required for enhancer activation
- **Indication / model:** SMARCB1-deficient rhabdoid cell lines with inducible SMARCB1 re-expression; ChIP-seq and ATAC-seq
- **Design:** SMARCB1 loss collapses BAF occupancy and remodelling at typical enhancers while residual complex persists at super-enhancers
- **Effect size:** strong — Re-expressing SMARCB1 restored BAF binding and accessibility at thousands of typical enhancers and at bivalent promoters. The residual SMARCB1-less complex retained super-enhancer occupancy, which explains why loss of this one subunit reprograms rather than abolishes BAF function.
- **Variance:** vs isogenic SMARCB1 re-expressing cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Isogenic rhabdoid systems only; no smooth-muscle or mesenchymal-lineage cells tested. Supports why complete INI1 protein loss, not a heterozygous variant call, is the biologically meaningful state.

Honma/Adachi (2017) — Cancer Sci

Reference:PMID 28741798 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EZH1/2 dual inhibitors (valemetostat chemical series) — antitumor profile and SWI/SNF correlation
- **Indication / model:** haematological and solid cancer cell-line panels plus DLBCL xenografts; 14-day rat tolerability

- **Design:** simultaneous inhibition of EZH1- and EZH2-containing PRC2 suppresses H3K27me3 more completely than EZH2-selective blockade
- **Effect size:** moderate — Dual EZH1/2 inhibition dropped H3K27me3 further and outperformed the EZH2-selective comparator in EZH2 gain-of-function DLBCL in vitro and in vivo. Across the solid-tumour panel the authors found no clear correlation between sensitivity and SWI/SNF mutation status, with few exceptions.
- **Variance:** vs EZH2-selective inhibitor as the head-to-head comparator; vehicle; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** The absent SWI/SNF-sensitivity correlation cuts against using a SMARCB1 call as the selection biomarker for this class. No sarcoma or smooth-muscle model appears in the panel, and the compounds are the chemical series behind valemetostat rather than the clinical agent itself.

Cuppens/Amant (2017) — Gynecol Oncol

Reference:PMID 28625393 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Uterine sarcoma patient-derived xenograft panel — engraftment and fidelity
- **Indication / model:** subcutaneous patient-derived xenografts from uterine sarcoma and carcinosarcoma in immunocompromised mice
- **Design:** not applicable — model characterisation
- **n:** 13 LMS implanted, 10 engrafted (77%); 2/7 carcinosarcoma (29%); 1 high-grade uterine sarcoma NOS
- **Effect size:** Leiomyosarcoma engrafted at 77% and the xenografts looked histologically like their source tumours, with desmin or h-caldesmon retained in 8 of 10. Copy-number concordance with the parent tumour ranged from 57.7% to 98.2%, and three LMS models clustered away from their originating tumour by RNA expression.
- **Variance:** vs paired original patient tumour for histology, copy-number and RNA comparison; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Subcutaneous rather than orthotopic, immunodeficient hosts, and the RNA drift in three of ten LMS models is the honest limit on how far a uLMS PDX result travels. This is close to the whole published uLMS PDX resource.

McGranahan/TRACERx Consortium (2017) — Cell

Reference:PMID 29107330 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Allele-specific HLA loss as an immune-escape route
- **Indication / model:** multiregion whole-exome sequencing of primary non-small-cell lung tumours (TRACERx) with an HLA-specific LOH caller
- **Design:** somatic loss of one HLA haplotype removes the restriction element without touching the antigen, under immune-editing selection
- **n:** LOH at the HLA locus in 40% of tumours analysed
- **Effect size:** strong — HLA LOH occurred in around 40% of tumours, was frequently subclonal and enriched in metastatic or later subclones, and associated with a higher neoantigen burden in the affected regions.
- **Variance:** vs HLA-intact regions within the same tumours; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Lung cancer under endogenous T-cell pressure, not sarcoma under engineered T cells. The relevance

is that germline HLA-A*02:01 typing does not guarantee the pulmonary metastases still present that allele.

Chang/Scheinberg (2017) — J Clin Invest

Reference:PMID 28628042 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRAME/HLA-A2 epitope density and the immunoproteasome
- **Indication / model:** PRAME-positive HLA-A2-positive human cancer and leukemia lines; mouse xenograft models of human leukemia; TCR-mimic antibody Pr20
- **Design:** the PRAME300-309 ALYVDSLFFL peptide is presented on HLA-A*02:01 at very low copy number, and immunoproteasome subunit beta5i induction after IFN-gamma raises the amount presented
- **Effect size:** moderate — Surface PRAME/HLA-A2 complexes were present at ultra-low density, and an afucosylated TCR-mimic antibody was nonetheless therapeutically active against leukemia xenografts. IFN-gamma markedly increased Pr20 binding in some tumours by shifting proteasome cleavage away from destroying the epitope.
- **Variance:** vs PRAME-negative and HLA-A2-negative lines; unmodified IgG; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Leukemia and melanoma models, and a TCR-mimic antibody rather than a TCR-T or ImmTAC. The transferable point is that PRAME protein positivity by IHC and PRAME epitope density on HLA-A2 are not the same quantity.

Allen/Bergers (2017) — Sci Transl Med

Reference:PMID 28404866 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Antiangiogenic plus anti-PD-L1 therapy and high endothelial venule formation
- **Indication / model:** mouse models of pancreatic neuroendocrine tumour, mammary carcinoma and glioblastoma
- **Design:** VEGF blockade normalises tumour vasculature and induces high endothelial venules, which in turn permit cytotoxic T-cell entry that checkpoint blockade can then sustain
- **Effect size:** strong — The combination prolonged survival and produced tumour regression in models where either agent alone did little, and the benefit tracked with intratumoural high endothelial venule formation and increased CTL infiltration. Anti-PD-L1 also sustained vessel normalisation, making the interaction reciprocal.
- **Variance:** vs single-agent anti-VEGFR2, single-agent anti-PD-L1, vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Carcinoma and neuroendocrine models, no sarcoma. This is the mechanistic ancestor of the PD-1 x VEGF bispecific rationale rather than evidence about ivonescimab itself.

—/— (2017)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** anlotinib (catequentinib, AL3818)
- **Indication / model:** advanced alveolar soft part sarcoma, leiomyosarcoma, synovial sarcoma (2L+); —
- **Design:** randomised phase 3 (APROMISS)
- **n:** 325

- **Effect size:** — — objective response rate (alveolar soft part sarcoma cohort)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Demetri/Patel (2016) — J Clin Oncol

Reference:PMID 26371143 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Trabectedin (ET743-SAR-3007 phase 3)
- **Indication / model:** metastatic liposarcoma or leiomyosarcoma after anthracycline and at least one further regimen (2L+); 518 patients; L-sarcomas only; leiomyosarcoma 73% of enrolment
- **Design:** randomised open-label phase 3 (NCT01343277)
- **n:** 518
- **Effect size:** 4.2 months — median PFS
- **Variance:** p=<0.001
- **Toxicities:** neutropenia G>=3 (37%); ALT elevation G>=3 (26%); thrombocytopenia G>=3 (17%); anemia G>=3 (14%); rhabdomyolysis G>=3 (1.2%); treatment-related death G5 (2.1%)
- **Case fit:** partial
- **Notes:** Basis of trabectedin's US approval in L-sarcomas: disease control without an OS difference. Per-term G3-4 rates from the PMC full text. If she receives trabectedin inside LMS-04, this later-line monotherapy row becomes less relevant to her own sequence.

Benson/van der Graaf (2016) — Gynecol Oncol

Reference:PMID 27012429 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pazopanib in uterine sarcoma (EORTC 62043/62072 retrospective)
- **Indication / model:** advanced uterine sarcoma (39 of 44 uterine LMS), heavily pretreated (2L+); 44 uterine sarcoma patients pooled from EORTC phase 2 (62043) and PALETTE (62072); 61% had >=2 prior chemotherapy lines; 84% high grade
- **Design:** retrospective analysis of two prospective trials
- **n:** 44
- **Effect size:** 11 % — partial response rate
- **Variance:** 95% CI 3.8–24.6
- **Toxicities:** No uterine-specific safety analysis; toxicity in the parent trials matches the PALETTE profile on the companion row.
- **Case fit:** partial
- **Notes:** Confirms pazopanib activity reaches uterine LMS at roughly the same modest level as other STS. AE fallback: abstract has no per-term data, no PMC record, publisher paywalled; safety is carried on the PALETTE row for the same drug and dose.

Schöffski/Patel (2016) — Lancet

Reference:PMID 26874885 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eribulin (phase 3 vs dacarbazine)
- **Indication / model:** advanced liposarcoma or leiomyosarcoma, >=2 prior regimens including an anthracycline

(2L+); 452 patients; LMS and liposarcoma randomised as independent strata
- **Design:** randomised open-label phase 3 (NCT01327885)
- **n:** 452
- **Effect size:** 13.5 months — median overall survival
- **Variance:** 95% CI 10.9–15.6; p=0.0169
- **Toxicities:** any adverse event G>=3 (152/226, 67%); death on treatment G5 (10/226, 4%)
- **Case fit:** partial
- **Notes:** The trial-level OS benefit was driven by the liposarcoma stratum (see the excluded Demetri 2017 subgroup row); the LMS subgroup did not share the OS advantage, so eribulin sits low among her later-line options despite the positive headline. Abstract reports only pooled G3+ rates; per-term table is in the full paper.

Reck Dos Santos/Cypel (2016) — Ann Thorac Surg

Reference:PMID 26952295 · **Type:** clinical · **Inclusion:** excluded — Preclinical porcine feasibility study, outside clinical-evidence scope. Read because the Toronto in vivo lung perfusion trial (NCT02811523) in the trial screen rests on it; no human efficacy publication exists as of August 2026. The trial's doxorubicin >450 mg cumulative-dose exclusion is the sequencing conflict with LMS-04 induction.

- **Intervention:** In vivo lung perfusion with doxorubicin (preclinical)
- **Notes:** Excluded: Preclinical porcine feasibility study, outside clinical-evidence scope. Read because the Toronto in vivo lung perfusion trial (NCT02811523) in the trial screen rests on it; no human efficacy publication exists as of August 2026. The trial's doxorubicin >450 mg cumulative-dose exclusion is the sequencing conflict with LMS-04 induction.

Mäkinen/Vahteristo (2016) — PLoS Genet

Reference:PMID 26891131 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Exome landscape of uterine leiomyosarcoma — TP53, ATRX, MED12
- **Indication / model:** whole-exome sequencing of uterine leiomyosarcoma specimens with matched normal DNA
- **Design:** defines which genes are recurrently hit in this specific histology
- **Effect size:** TP53, ATRX and MED12 were the recurrently mutated genes in uterine leiomyosarcoma. SMARCB1 was not among the recurrent events.
- **Variance:** vs matched germline DNA; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive sequencing with no therapeutic arm. Useful mainly as the denominator against which the plasma SMARCB1 call should be judged.

Deeg/Rippe (2016) — Front Oncol

Reference:PMID 27602331 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ALT cells are not generally hypersensitive to ATR inhibition — failed replication
- **Indication / model:** panel of ALT-positive and telomerase-positive cell lines including U2OS and ALT-positive glioma lines
- **Design:** direct test of whether the ALT phenotype itself confers ATR dependency
- **Effect size:** negative — Across a broader panel and several ATR inhibitors, ALT-positive lines showed no

general hypersensitivity relative to telomerase-positive lines.
- **Variance:** vs telomerase-positive lines; multiple ATR inhibitors; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Contradicts Flynn 2015. In-vitro only, but the negative result is the reason ALT status alone should not be treated as an ATR-inhibitor selection biomarker in this case.

—/— (2016)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** dostarlimab
- **Indication / model:** advanced solid tumors, including a mismatch-repair-deficient basket cohort (2L+); mismatch-repair deficiency / MSI-high
- **Design:** phase 1 dose escalation with expansion cohorts (GARNET)
- **n:** 730
- **Effect size:** — — treatment-emergent adverse events and objective response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2016)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** in vivo lung perfusion with doxorubicin, plus metastasectomy
- **Indication / model:** bilateral pulmonary metastases from bone or soft-tissue sarcoma (n/a (locoregional)); —
- **Design:** phase 1 dose escalation of in vivo lung perfusion
- **n:** 17
- **Effect size:** — — acute lung injury after in vivo lung perfusion
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Hensley/Michael (2015) — J Clin Oncol

Reference:PMID 25713428 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + docetaxel +/- bevacizumab (GOG-0250)
- **Indication / model:** chemotherapy-naive metastatic unresectable uterine leiomyosarcoma (1L); 107 of a planned 130 patients; accrual stopped early for futility
- **Design:** randomised double-blind placebo-controlled phase 3 (GOG-0250)
- **n:** 107
- **Effect size:** 1.12 HR — median PFS (bevacizumab did not improve it)
- **Variance:** p=0.58
- **Toxicities:** neutropenia G>=3 (12/52, 23%); thrombocytopenia G>=3 (19/52, 37%); anemia G>=3 (7/52, 13%); hypertension G3 (4/52, 8%); thromboembolic event G>=3 (5/52, 10%)
- **Case fit:** strong

- **Notes:** The uterine-LMS-specific test of adding plain VEGF blockade to first-line chemotherapy, and it failed. Directly relevant when weighing the PD-1xVEGF bispecific rationale for the ivonescimab trial seat: anti-VEGF alone bought nothing in this exact disease. Per-term AE rates from the PMC full text (Table 2, n=103 toxicity-evaluable).

Kim/Roberts (2015) — Nat Med

Reference:PMID 26552009 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EZH2 dependency in SWI/SNF-mutant cancers is largely non-catalytic
- **Indication / model:** panel of cancer cell lines and xenografts mutant for ARID1A, PBRM1 or SMARCA4; shRNA/CRISPR versus enzymatic inhibitor
- **Design:** EZH2 supports PRC2 complex stability independently of its methyltransferase activity, so genetic depletion and enzymatic inhibition are not equivalent
- **Effect size:** moderate — EZH2 was essential across the SWI/SNF-mutant lines tested, but the dependency ran mostly through a non-catalytic stabilising role, with only partial reliance on methyltransferase activity. A co-occurring Ras-pathway mutation abrogated the dependency altogether.
- **Variance:** vs SWI/SNF-wild-type lines; catalytically dead and stabilisation-competent EZH2 rescue constructs; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** The mutant subunits tested were ARID1A, PBRM1 and SMARCA4, not SMARCB1. The practical warning for this case is that an enzymatic EZH2 inhibitor may under-deliver even where the genetic dependency is real.

Smith/Houghton (2015) — Clin Cancer Res

Reference:PMID 25500058 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP inhibitor plus temozolomide across paediatric solid-tumour xenografts
- **Indication / model:** Pediatric Preclinical Testing Program xenograft panel including Ewing sarcoma, rhabdomyosarcoma and osteosarcoma
- **Design:** PARP trapping potentiating temozolomide-induced base damage in vivo
- **Effect size:** moderate — Talazoparib alone was largely inactive, while the combination with temozolomide produced objective regressions in a subset of xenografts, most consistently Ewing sarcoma. Myelosuppression forced temozolomide dose reduction in the combination.
- **Variance:** vs talazoparib alone, temozolomide alone, vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No leiomyosarcoma model in the panel, and the responsive histology (Ewing) has a distinct DNA-damage biology. The haematologic toxicity signal is the part that carries directly into clinic.

Liau/Yang (2015) — Am J Surg Pathol

Reference:PMID 25229770 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ALT prevalence and ATRX loss in leiomyosarcoma
- **Indication / model:** 92 human leiomyosarcomas (uterine, retroperitoneal and other sites) by telomere FISH,

ATRX/DAXX IHC and TERT promoter sequencing
- **Design:** ATRX inactivation permits the recombination-based telomere maintenance that defines ALT
- **n:** n=92 tumours
- **Effect size:** moderate — ALT was present in 59% of leiomyosarcomas and ATRX expression was lost in 33%, with all but two ATRX-deficient tumours being ALT-positive. Both features went with epithelioid or pleomorphic morphology, necrosis and worse outcome.
- **Variance:** vs ALT-negative tumours within the same series; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Human tissue survey with no therapeutic arm. Sets a realistic pre-test probability that this tumour is ALT-positive, which matters only if an ALT-directed strategy has a real preclinical basis.

Flynn/Zou (2015) — Science

Reference:PMID 25593184 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibition in ALT-positive cells — the original claim
- **Indication / model:** ALT-positive and telomerase-positive human cancer cell lines, including ATRX-deficient lines
- **Design:** ALT depends on ATR-coordinated homologous recombination at telomeres, so ATR inhibition disrupts telomere maintenance selectively
- **Effect size:** strong — ALT-positive lines were selectively killed by ATR inhibitors while telomerase-positive lines were spared, and ATRX loss was linked to the ALT phenotype.
- **Variance:** vs isogenic and matched telomerase-positive lines; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** In-vitro only, and the selectivity claim failed to replicate in later work — read alongside Deeg 2016 and Goncalves 2020 rather than on its own.

Judson/van der Graaf (2014) — Lancet Oncol

Reference:PMID 24618336 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Doxorubicin + ifosfamide (EORTC 62012)
- **Indication / model:** advanced or metastatic high-grade soft-tissue sarcoma, first line (1L); age 18-60, WHO PS 0-1; all STS histologies, not LMS- or uterine-specific
- **Design:** randomised controlled phase 3 (EORTC 62012, NCT00061984)
- **n:** 455
- **Effect size:** 14.3 months — median overall survival (primary endpoint, not met)
- **Variance:** 95% CI 12.5–16.5; p=0.076
- **Toxicities:** febrile neutropenia G>=3 (103/224, 46%); leucopenia G>=3 (97/224, 43%); neutropenia G>=3 (93/224, 42%); anemia G>=3 (78/224, 35%); thrombocytopenia G>=3 (75/224, 33%)
- **Case fit:** partial
- **Notes:** Doubles the response rate without an OS win, at real toxicity cost; the authors themselves restrict it to situations where tumour shrinkage is the goal. Age cap 60 means she would have been eligible. All-comer STS, so LMS-specific benefit is not established here.

Murai/Pommier (2014) — J Pharmacol Exp Ther

Reference:PMID 24650937 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP trapping as the reason olaparib pairs with temozolomide
- **Indication / model:** isogenic DT40 and human cell lines with defined DNA-repair deletions; PARP inhibitors compared for trapping versus catalytic potency
- **Design:** temozolomide-induced N-methylpurine lesions generate base-excision-repair intermediates that trapped PARP-DNA complexes convert into cytotoxic replication blocks
- **Effect size:** strong — Synergy with temozolomide tracked with a drug's ability to trap PARP on DNA rather than with catalytic inhibition, so trapping-potent inhibitors combined far better than veliparib. Camptothecin combinations followed the same rule.
- **Variance:** vs PARP1/2-null cells; catalytic-only inhibitor (veliparib) comparator; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Engineered cell lines rather than sarcoma models. Explains why the olaparib-plus-temozolomide regimen is a mechanistic pairing rather than an empirical one.

Knutson/Keilhack (2013) — Proc Natl Acad Sci U S A

Reference:PMID 23620515 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EPZ-6438 (tazemetostat) in SMARCB1-deleted rhabdoid tumours
- **Indication / model:** SMARCB1-deleted malignant rhabdoid tumour cell lines and subcutaneous xenografts in mice
- **Design:** selective inhibition of EZH2 methyltransferase activity, lowering H3K27me3 and releasing differentiation programs in SMARCB1-null cells
- **Effect size:** strong — Apoptosis and differentiation followed EZH2 inhibition specifically in SMARCB1-deleted rhabdoid lines. Xenografts regressed dose-dependently with matching loss of intratumoural H3K27me3, and tumours did not regrow after dosing stopped.
- **Variance:** vs vehicle-treated xenografts; SMARCB1-wild-type lines as the specificity comparator; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** This is the origin of the whole SMARCB1-to-EZH2 therapeutic logic, and every model in it is a paediatric rhabdoid tumour with biallelic SMARCB1 deletion. All authors but one were Epizyme employees.

Cameron/Jakobsen (2013) — Sci Transl Med

Reference:PMID 23926201 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Titin cross-reactivity of an affinity-enhanced MAGE-A3 TCR — the class safety lesson
- **Indication / model:** in-vitro functional analysis of an affinity-enhanced HLA-A*01-restricted MAGE-A3 TCR; post-hoc amino-acid scanning and beating-cardiomyocyte cultures
- **Design:** affinity enhancement broadens peptide tolerance, so a structurally similar peptide from a muscle protein becomes a target
- **Effect size:** strong — Conventional preclinical screening found no off-target activity, yet the engineered T cells caused fatal cardiac toxicity in patients. Amino-acid scanning afterwards identified the titin peptide ESDPIVAQY

as the cross-reactive target.
- **Variance:** vs pre-SAE preclinical panels that had returned clean; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** A different TCR, a different antigen and a different HLA allele from the MAGE-A4 A*02:01 products in scope here. Logged because it is why the peptide-scanning and primary-cell work in the MAGE-A4 package matters, and because the off-target was a sarcomeric muscle protein.

Wang/Zhang (2013) — PLoS One

Reference:PMID 23940627 · **Type:** preclinical · **Inclusion:** excluded — Osteosarcoma-only expression survey, superseded for this case by the 153-patient soft-tissue sarcoma series (Lynch 2024) that includes leiomyosarcoma and reports vessel staining as well.

- **Intervention:** B7-H3 overexpression in osteosarcoma
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Osteosarcoma-only expression survey, superseded for this case by the 153-patient soft-tissue sarcoma series (Lynch 2024) that includes leiomyosarcoma and reports vessel staining as well.

Germano/Allavena (2013) — Cancer Cell

Reference:PMID 23410977 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Trabectedin's selective depletion of mononuclear phagocytes
- **Indication / model:** four mouse tumour models with myeloid-cell transfer and depletion experiments; trabectedin-resistant tumour cells; blood and tumour from treated patients
- **Design:** caspase-8-dependent apoptosis restricted to mononuclear phagocytes, with selectivity over neutrophils and lymphocytes set by differential signalling and decoy TRAIL receptor expression
- **Effect size:** strong — Trabectedin depleted monocytes and macrophages from blood, spleen and tumour across four mouse models, with angiogenesis falling alongside. Transfer and depletion experiments using trabectedin-resistant tumour cells established that killing mononuclear phagocytes is a real component of the antitumour effect, and monocyte depletion was reproduced in treated patients.
- **Variance:** vs untreated tumour-bearing mice; trabectedin-resistant tumour cells to separate direct from myeloid effects; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** None of the four mouse models is uterine leiomyosarcoma. Directly relevant here because the metastatic uLMS niche is macrophage-rich, which makes the planned first-line agent partly an immune-modulating one.

Pang/Neefjes (2013) — Nat Commun

Reference:PMID 23715267 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Histone eviction as a second mechanism of anthracycline action
- **Indication / model:** human cancer cell lines, mouse tissues including heart, and primary AML blasts from patients
- **Design:** anthracyclines evict histones, including H2AX, from open chromatin independently of

double-strand-break formation, which deregulates transcription and blunts the damage response
- **Effect size:** strong — Doxorubicin evicted histones from open chromatin whether or not it induced double-strand breaks, and H2AX eviction was accompanied by attenuated DNA repair. The same eviction happened in heart tissue and could kill topoisomerase-negative AML blasts.
- **Variance:** vs etoposide and doxorubicin variants that break DNA without evicting histones; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No sarcoma model. Matters here for two reasons: it explains anthracycline activity in slowly proliferating tumours, and it links the same mechanism to cardiac tissue in a patient whose baseline LVEF has not yet been measured.

van der Graaf/Hohenberger (2012) — Lancet

Reference:PMID 22595799 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pazopanib (PALETTE)
- **Indication / model:** metastatic non-adipocytic soft-tissue sarcoma after standard chemotherapy (2L+); 369 patients, angiogenesis-inhibitor-naive; leiomyosarcoma was the largest histology group
- **Design:** randomised double-blind placebo-controlled phase 3 (PALETTE, NCT00753688)
- **n:** 369
- **Effect size:** 4.6 months — median PFS
- **Variance:** 95% CI 3.7–4.8; p=<0.0001
- **Toxicities:** fatigue (65%); diarrhoea (58%); nausea (54%); weight loss (48%); hypertension (41%)
- **Case fit:** partial
- **Notes:** Approval trial for pazopanib in pretreated STS. PFS gain of 3 months, no OS gain. The uterine-sarcoma-specific read is on the Benson retrospective row.

García-Del-Muro/Buesa (2011) — J Clin Oncol

Reference:PMID 21606430 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + dacarbazine (Spanish GEIS randomised phase 2)
- **Indication / model:** previously treated advanced soft-tissue sarcoma (2L+); 113 patients, mixed STS histologies including LMS; not uterine-specific
- **Design:** randomised phase 2 vs dacarbazine
- **n:** 113
- **Effect size:** 56 % progression-free at 3 months — progression-free rate at 3 months (primary endpoint)
- **Variance:** p=0.001
- **Toxicities:** Well tolerated; granulocytopenia the main serious event, febrile neutropenia uncommon, discontinuation for toxicity rare. Abstract gives no per-term percentages.
- **Case fit:** weak
- **Notes:** A gemcitabine-based fallback with an OS signal in a small randomised phase 2; mixed histologies, so LMS-specific benefit is inferred rather than shown. AE fallback: abstract qualitative only, no PMC record, publisher paywalled.

Burt/Jaklitsch (2011) — Ann Thorac Surg

Reference:PMID 21867989 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Repeated pulmonary metastasectomy for leiomyosarcoma
- **Indication / model:** pulmonary metastases from sarcoma; LMS subset n=31 of 82 (any); single-institution series over 15 years; LMS patients presented with fewer metastases (mean 1.9) and were mostly female
- **Design:** retrospective single-institution cohort
- **n:** 82
- **Effect size:** 70 months — median OS after metastasectomy, LMS vs other sarcoma
- **Variance:** p=0.049
- **Toxicities:** Surgical morbidity not summarised in the abstract; disease-free intervals before and after first metastasectomy were the independent survival predictors.
- **Case fit:** partial
- **Notes:** LMS lung metastases behave more indolently than other sarcomas after resection, and repeat operations extend survival in selected patients. Selection bias is heavy: these were operable, often oligometastatic patients, which she currently is not.

Wilson/Roberts (2010) — Cancer Cell

Reference:PMID 20951942 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Snf5 loss drives EZH2 dependence — genetic proof
- **Indication / model:** conditional Snf5 mouse crossed to conditional Ezh2; SNF5-deficient MEFs and human rhabdoid tumours
- **Design:** SNF5 loss raises EZH2 and drives H3K27me3 across Polycomb targets; the two complexes act antagonistically on stem-cell programs
- **Effect size:** strong — Polycomb targets were broadly trimethylated and repressed in SNF5-deficient cells, and inactivating Ezh2 genetically blocked tumour formation driven by Snf5 loss. The dependency is genetic, not merely correlative.
- **Variance:** vs Snf5-loss animals with intact Ezh2; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Whole rationale rests on complete Snf5 loss in a mouse rhabdoid context. Nothing here tests a smooth-muscle lineage or a heterozygous truncating allele. An erratum was published (Cancer Cell 2011;19:153).

Pérot/Aurias (2009) — Cancer Res

Reference:PMID 19276386 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MYOCD amplification as a lineage dependency in leiomyosarcoma
- **Indication / model:** human retroperitoneal leiomyosarcoma series with siRNA knockdown in LMS cell lines
- **Design:** myocardin amplification sustains the smooth-muscle transcriptional program that defines the lineage
- **Effect size:** moderate — MYOCD was amplified and overexpressed in most well-differentiated retroperitoneal leiomyosarcomas, and knocking it down reduced smooth-muscle marker expression and proliferation.
- **Variance:** vs non-targeting siRNA; non-amplified tumours; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Retroperitoneal rather than uterine LMS, and no drug exists against myocardin. Included because lineage dependency is the alternative frame to the borrowed epigenetic one.

Xing/Orsulic (2009) — Cancer Res

Reference:PMID 19843854 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Conditional p53/BRCA1 deletion in the murine uterus — a uLMS model and human BRCA1 loss
- **Indication / model:** Amhr2-Cre conditional p53 and p53/BRCA1 deleted mice; human uterine leiomyosarcoma specimens for BRCA1 protein and promoter methylation
- **Design:** p53 loss initiates uterine smooth-muscle tumours and concurrent BRCA1 loss accelerates them, with promoter methylation as the human silencing route
- **Effect size:** moderate — Conditional p53 deletion produced uterine tumours resembling human uLMS, and adding BRCA1 deletion accelerated progression. BRCA1 protein was absent in 29% of human uLMS, most plausibly through promoter methylation.
- **Variance:** vs p53-deleted-only animals; BRCA1-intact human tumours; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Genetically engineered mouse tumours are not identical to human uLMS, and the 29% BRCA1-negative figure is IHC-based rather than an HRD score. Still the clearest lineage-matched argument that an HR-directed branch is worth testing on tissue.

Hensley/Rose (2008) — Gynecol Oncol

Reference:PMID 18534250 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + docetaxel, first-line uterine LMS (GOG phase 2)
- **Indication / model:** metastatic uterine leiomyosarcoma, chemotherapy-naive (1L); 42 women with advanced uLMS, no prior chemotherapy; 39 evaluable for response
- **Design:** single-arm phase 2 (Gynecologic Oncology Group)
- **n:** 42
- **Effect size:** 35.8 % — objective response rate
- **Variance:** 95% CI 23.5–49.6
- **Toxicities:** neutropenia G3 (n=42, 5%); neutropenia G4 (n=42, 12%); anemia G3 (n=42, 24%); thrombocytopenia G3 (n=42, 9.5%); fatigue G3 (n=42, 17%); pulmonary toxicity G4 (1/42, 2.4%)
- **Case fit:** strong
- **Notes:** The uterine-specific first-line benchmark for gem-doce: response in a third of patients but short mPFS. Read alongside GeDDiS before treating the 35.8% as superiority over doxorubicin; single-arm response rates in uLMS have repeatedly outrun randomised comparisons.

Aune/Pommier (2008) — Clin Cancer Res

Reference:PMID 18927284 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Transcription-coupled repair and VHL-dependent RNA polymerase II degradation after trabectedin
- **Indication / model:** human cell lines with defined defects in transcription-coupled nucleotide excision repair and in VHL
- **Design:** trabectedin-DNA adducts stall elongating RNA polymerase II, and degradation of the stalled polymerase requires both transcription-coupled NER and VHL
- **Effect size:** strong — Trabectedin caused rapid RNA polymerase II degradation that depended on both TC-NER and VHL, tying cytotoxicity to transcription rather than to replication alone.

- **Variance:** vs TC-NER-proficient and VHL-proficient isogenic lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Isogenic human cell lines, no tumour model. Explains why trabectedin is a transcription-targeted drug and why repair-pathway status, not proliferation rate, is where a predictive biomarker would sit.

Herrero/Moreno (2006) — Cancer Res

Reference:PMID 16912194 · **Type:** preclinical · **Inclusion:** excluded — Mechanism worked out entirely in fission yeast with rad13 mutants; the mammalian version of the same NER-dependence and HR-repair claim is covered by Damia 2001 and Aune 2008 without the species gap.

- **Intervention:** NER and homologous-recombination cross-talk in trabectedin's mechanism
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Mechanism worked out entirely in fission yeast with rad13 mutants; the mammalian version of the same NER-dependence and HR-repair claim is covered by Damia 2001 and Aune 2008 without the species gap.

Epping/Bernards (2005) — Cell

Reference:PMID 16179254 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRAME as a dominant repressor of retinoic-acid receptor signalling
- **Indication / model:** human melanoma and other cancer lines with PRAME knockdown; in-vitro and in-vivo proliferation assays
- **Design:** PRAME binds liganded RAR and recruits Polycomb proteins to RAR target promoters, blocking retinoic-acid-induced differentiation and growth arrest
- **Effect size:** moderate — PRAME knockdown restored RAR signalling in RA-resistant melanoma and reinstated sensitivity to retinoic acid in vitro and in vivo.
- **Variance:** vs control shRNA; PRAME-low lines; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Melanoma, no sarcoma model. Relevant to the antigen-stability question: PRAME appears to be functionally selected for in the tumour rather than being inert cargo, which argues against easy antigen-loss escape but does not prove it.

Talbot/Taub (2003) — Cancer

Reference:PMID 14584078 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Temozolomide in soft-tissue sarcoma (LMS-enriched activity)
- **Indication / model:** unresectable or metastatic soft-tissue sarcoma, previously treated (2L+); 25 evaluable patients; all responders had uterine or non-uterine LMS; LMS subset n=11
- **Design:** single-arm phase 2
- **n:** 25
- **Effect size:** 8 % — objective response rate
- **Toxicities:** nausea G3 (1/25, 4%); anemia G3 (1/25, 4%); fatigue G3 (1/25, 4%)
- **Case fit:** partial

- **Notes:** Modest single-agent activity concentrated in LMS; today temozolomide's real interest for her is as the partner in olaparib-temozolomide rather than as monotherapy.

Meco/Riccardi (2003) — Cancer Chemother Pharmacol

Reference:PMID 12783202 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ET-743 combined with doxorubicin in sarcoma models
- **Indication / model:** TE-671 rhabdomyosarcoma xenografts in nude mice; UV2237M murine fibrosarcoma and its doxorubicin-resistant Pgp-overexpressing subline
- **Design:** a transcription-targeted minor-groove binder combined with a topoisomerase II poison, hitting non-overlapping lesions
- **Effect size:** moderate — Isobologram analysis showed an additive interaction with combination indices slightly below 1. In TE-671 xenografts each drug alone was marginal (log cell kill 0.13 and 0.33) while the combination reached 0.85 to 1.12, best when ET-743 preceded doxorubicin by an hour, and no pharmacokinetic interaction was found.
- **Variance:** vs single-agent ET-743, single-agent doxorubicin, untreated controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Pre-dates the LMS-04 regimen by nearly two decades and neither model is leiomyosarcoma. The interaction was additive rather than synergistic, and in the fibrosarcoma lung-metastasis arm the combination did not beat either drug alone.

Hensley/Spriggs (2002) — J Clin Oncol

Reference:PMID 12065559 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + docetaxel in leiomyosarcoma (first phase 2, mixed line)
- **Indication / model:** unresectable leiomyosarcoma, uterine-dominant (29 of 34 uterine) (any); 0-2 prior regimens; 16 of 34 had progressed on doxorubicin, 18 chemotherapy-naive
- **Design:** single-arm phase 2
- **n:** 34
- **Effect size:** 53 % — overall response rate
- **Variance:** 95% CI 35–70
- **Toxicities:** neutropenia G3 (n=34, 15%); neutropenia G4 (n=34, 6%); thrombocytopenia G3 (n=34, 26%); febrile neutropenia G>=3 (n=34, 6%)
- **Case fit:** partial
- **Notes:** The study that established gem-doce in LMS, and the source of the often-quoted 53% response rate. Activity held in doxorubicin-pretreated patients, which is why this regimen remains her most likely second-line chemotherapy after LMS-04.

Roberts/Orkin (2002) — Cancer Cell

Reference:PMID 12450796 · **Type:** preclinical · **Inclusion:** excluded — The tumour spectrum after conditional biallelic Snf5 loss was mature CD8+ T-cell lymphoma with only rare rhabdoid tumours, so the model speaks to lymphoid rather than mesenchymal transformation; the allele configuration is also biallelic, which is not the patient's called genotype.

- **Intervention:** Conditional Snf5 inversion — tumour spectrum after biallelic loss
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: The tumour spectrum after conditional biallelic Snf5 loss was mature CD8+ T-cell lymphoma with only rare rhabdoid tumours, so the model speaks to lymphoid rather than mesenchymal transformation; the allele configuration is also biallelic, which is not the patient's called genotype.

Guidi/Jones (2001) — Mol Cell Biol

Reference:PMID 11313485 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Ini1 heterozygous mice — second-hit requirement (LOH) for tumour formation
- **Indication / model:** germline Ini1 knockout mouse; tumours from heterozygotes genotyped at the Ini1 locus
- **Design:** loss of the remaining wild-type allele as the rate-limiting event downstream of a single inactivating hit
- **n:** ~15% of heterozygous mice developed tumours
- **Effect size:** moderate — About 15% of Ini1-heterozygous mice developed mostly undifferentiated sarcomas, and tumour formation went along with loss of heterozygosity at the Ini1 locus. Ini1-null embryos died between 3.5 and 5.5 days post-coitum.
- **Variance:** vs wild-type and untransformed heterozygous tissue; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** This is the paper that makes the mono- versus bi-allelic distinction concrete: a single truncating allele was not the transformed state, LOH was. Mouse germline model, undifferentiated sarcoma rather than smooth muscle.

Damia/D'Incalci (2001) — Int J Cancer

Reference:PMID 11304695 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ET-743 activity across cells with defined DNA-repair defects
- **Indication / model:** yeast and mammalian cell panels with defined NER, mismatch-repair, DNA-PK and ATM deficiencies
- **Design:** NER machinery converts the minor-groove adduct into a lethal lesion, while double-strand-break repair capacity determines how well the cell survives it
- **Effect size:** moderate — NER-deficient lines were 2- to 8-fold less sensitive to ET-743 and restoring NER re-sensitised them, the opposite of the usual pattern for DNA-damaging drugs. Cells lacking DNA-PK or ATM were more sensitive, and mismatch-repair status made no difference.
- **Variance:** vs repair-proficient parental lines and complemented revertants; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Engineered cell systems from 2001, no tumour model. The closest thing to a mechanistic biomarker of trabectedin benefit: intact NER plus impaired double-strand-break repair, which is the profile an HR-deficient uLMS would have.

Roberts/Orkin (2000) — Proc Natl Acad Sci U S A

Reference:PMID 11095756 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Snf5 heterozygous mice — allele dosage and rhabdoid tumour onset

- **Indication / model:** germline Snf5 knockout mouse (heterozygous and homozygous)
- **Design:** genetic test of how many SMARCB1 alleles must be lost before the rhabdoid phenotype appears
- **Effect size:** strong — Homozygous Snf5 null embryos died by embryonic day 7, while heterozygotes were born normal and only later developed focal rhabdoid-like tumours from about 5 weeks of age. One inactivated allele was therefore compatible with normal tissue, and tumours arose focally rather than across the whole heterozygous animal.
- **Variance:** vs wild-type littermates; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Mouse germline model of a paediatric rhabdoid phenotype; the tumours arise from first-branchial-arch soft tissue, not myometrium. Bears on this case only through the allele-dosage question.

Pastorino/Putnam (1997) — J Thorac Cardiovasc Surg

Reference:PMID 9011700 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pulmonary metastasectomy (International Registry of Lung Metastases)
- **Indication / model:** lung metastases from any primary; sarcoma 2173 of 5206 cases (any); 5206 metastasectomies from 18 thoracic units; prognostic model built on resectability, disease-free interval, and number of lesions
- **Design:** international multicentre registry
- **n:** 5206
- **Effect size:** 36 % 5-year OS — 5-year overall survival after complete resection
- **Toxicities:** Operative morbidity and mortality not reported in the abstract; the registry's message is selection: completeness of resection, DFI and lesion count drive outcome.
- **Case fit:** partial
- **Notes:** The selection criteria that still govern metastasectomy decisions. Her roughly 23-month interval from primary resection to metastatic diagnosis sits in the favourable DFI band, but bilateral multifocal infiltrative disease argues against completeness of resection today; this becomes live only after systemic control.

De Smet/Boon (1996) — Proc Natl Acad Sci U S A

Reference:PMID 8692960 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Promoter demethylation as the switch that turns on MAGE genes
- **Indication / model:** 20 human tumour cell lines plus primary fibroblasts; CpG methylation mapping and 5-aza-2'-deoxycytidine treatment
- **Design:** CpG demethylation at Ets-binding sites in the MAGE-1 promoter permits transcription-factor binding, downstream of genome-wide demethylation in the tumour
- **Effect size:** strong — MAGE-1 promoter CpGs were demethylated in expressing lines and methylated in non-expressing ones, and 5-aza-2'-deoxycytidine switched the gene on even in normal fibroblasts. Overall CpG methylation was inversely correlated with MAGE-1 expression across the 20 lines.
- **Variance:** vs MAGE-1-negative lines; untreated fibroblasts; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** MAGE-1 rather than MAGE-A4, and the work predates modern epigenomics. It is the reason cancer/testis antigen expression is patchy and epigenetically labile rather than a fixed lineage property — which is the honest answer to whether expression is stable enough for a TCR-T strategy.